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!pip install faker --quiet

import random
import re
import pandas as pd
from faker import Faker
from typing import List, Tuple, Set

fake = Faker()
Faker.seed(42)
random.seed(42)

JOB_KEYWORDS = {
    "Software Engineer": [
        "Java", "C++", "Python", "Go", "Rust", "Git", "System Design", "Algorithms",
        "Data Structures", "Docker", "Kubernetes", "CI/CD", "Microservices", "REST APIs",
        "HTTP", "Multithreading", "Concurrency", "OAuth", "PostgreSQL", "MySQL", "NoSQL",
        "Testing", "Unit Tests", "TDD", "Design Patterns", "Scalable Systems"
    ],
    "Product Manager": [
        "Roadmap", "Stakeholder Management", "Analytics", "Agile", "Scrum", "A/B Testing",
        "Wireframes", "User Stories", "OKRs", "KPIs", "Prioritization", "Product Strategy",
        "Go-to-market", "Market Research", "Customer Interviews", "Feature Scoping",
        "Revenue", "Monetization", "Product Metrics", "User Feedback", "Competitive Analysis"
    ],
    "Designer": [
        "Figma", "Sketch", "Adobe XD", "Photoshop", "Illustrator", "InDesign", "Prototyping",
        "UX Research", "User Testing", "Information Architecture", "Visual Design",
        "Interaction Design", "Design Systems", "Accessibility", "Typography",
        "Branding", "Wireframing", "User Flows", "High-fidelity", "Responsive Design"
    ]
}
ALL_CATEGORIES = list(JOB_KEYWORDS.keys())

def make_email(name: str) -> str:
    username = re.sub(r'^[a-z0-9]', '.', name.lower().replace(" ", "."))
    username = re.sub(r'\.+', '.', username).strip('.')
    domain = random.choice(["gmail.com", "hotmail.com", "example.com", "outlook.com", "icloud.com"])
    return f"{username}.{random.randint(1,9999)}@{domain}"

def truncate_to_word_limit(text: str, max_words: int = 80) -> str:
    words = text.split()
    if len(words) <= max_words:
        return text
    return " ".join(words[:max_words])

def pick_non_target_keywords(target: str, k: int) -> List[str]:
    others = [kw for cat, kws in JOB_KEYWORDS.items() if cat != target for kw in kws]
    if len(others) == 0:
        others = OTHER_PHRASES
    return random.sample(others, min(k, len(others)))

SUMMARY_TEMPLATES = [
    "Experienced {} with a strong background in {} and {}.",
    "{} specializing in {} with experience delivering {}.",
    "Practical {} skilled in {} and {}.",
    "{} with {} years of experience in {} and {}.",
    "Passionate {} focusing on {} and {} to deliver product impact.",
    "Research-oriented {} experienced in {} and {} for applied outcomes.",
    "Creative {} who leverages {} and {} to solve user problems.",
    "Results-driven {} with expertise in {} and a passion for {}.",
    "Dynamic and motivated {} adept at {} and {}.",
    "Seeking to apply my {} years of experience in {} to a challenging new role.",
    "A {} with a proven track record in {} and {}.",
]

WORK_TEMPLATES = [
    "Developed {} to improve {}.",
    "Led a team that implemented {} using {}.",
    "Conducted {} and reported findings in {}.",
    "Built production-ready {} and integrated with {}.",
    "Designed {} focused on {}.",
    "Performed {} for product improvements and {}.",
]

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"Managed {} and coordinated with {} stakeholders.",
"Architected {} to scale {} across services.",
"Implemented {} following {} best practices.",
"Optimized {} achieving {} improvement in {}.",
"Designed experiments on {} and analyzed {}.",
"Prototyped {} and tested with {} users.",
"Authored {} submitted to {}.",
"Automated {} in the {} pipeline.",
# New additions
"Collaborated with {} teams to deliver {} on schedule.",
"Analyzed {} to identify key opportunities for {}.",
"Drove the development of {} from concept to launch.",
"Overhauled the existing {} to increase {}.",
"Mentored junior team members in {} and {}.",
]

# New template category for skills
SKILL_TEMPLATES = [
    "Core competencies include {}, {}, and {}.",
    "Proficient in: {}, {}, {}.",
    "Expertise in {} and {}."
]

BULLET_ACTIONS = [
    "Improved", "Reduced", "Increased", "Automated", "Published", "Designed",
    "Implemented", "Measured", "Validated", "Scaled", "Refactored",
    "Pioneered", "Overhauled", "Streamlined", "Launched", "Negotiated"
]

BULLET_METRICS = [
    "performance", "latency", "accuracy", "user conversions", "throughput",
    "citation count", "engagement", "retention", "test coverage",
    "customer satisfaction", "system uptime", "revenue growth", "cost savings"
]

OTHER_PHRASES = [
    "cross-functional teams", "scalable systems", "large datasets", "user research",
    "experimental setups", "cloud infrastructure", "CI pipelines", "data pipelines",
    "mobile platforms", "web platforms", "stakeholder feedback", "production metrics",
    "end-to-end solutions", "industry best practices", "agile environments", "data-driven insights",
    "user-centric design", "technical specifications", "market trends"
]

def fill_template_with_args(tmpl: str, args: List[str]) -> str:
    n_placeholders = tmpl.count("{}")

    while len(args) < n_placeholders:
        args.append(random.choice(sum(JOB_KEYWORDS.values(), []) + OTHER_PHRASES))

    return tmpl.format(*args[:n_placeholders])

def generate_resume_text(
    target_category: str,
    positive: bool,
    max_words: int = 80,
    min_sentences: int = 3,
    max_sentences: int = 6
) -> Tuple[str, List[str]]:
    sentences = []
    num_sentences = random.randint(min_sentences, max_sentences)

    if positive:
        n_kw = random.randint(2, 5)
        kws = random.sample(JOB_KEYWORDS[target_category], min(n_kw, len(JOB_KEYWORDS[target_category])))
    else:
        kws = pick_non_target_keywords(target_category, random.randint(2, 4))

    keywords_used = kws.copy()

    for i in range(num_sentences):
        if i == 0:
            tmpl = random.choice(SUMMARY_TEMPLATES)
            k1 = random.choice(kws)
            k2 = random.choice(sum(JOB_KEYWORDS.values(), []) + OTHER_PHRASES)

            if "years" in tmpl:
                years = f"{random.randint(1,12)}+ years"

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        sentence = fill_template_with_args(tmpl, [target_category, years, k1])
    else:
        sentence = fill_template_with_args(tmpl, [target_category, k1, k2])

elif random.random() < 0.25:
    tmpl = random.choice(SKILL_TEMPLATES)
    skill_args = random.sample(kws, min(len(kws), tmpl.count("{}")))
    sentence = fill_template_with_args(tmpl, skill_args)

else:
    tmpl = random.choice(WORK_TEMPLATES)
    arg1 = random.choice(kws)
    arg2 = random.choice(OTHER_PHRASES + [random.choice(sum(JOB_KEYWORDS.values(), []))])
    arg3 = random.choice(OTHER_PHRASES + [random.choice(sum(JOB_KEYWORDS.values(), []))])
    sentence = fill_template_with_args(tmpl, [arg1, arg2, arg3])

sentences.append(sentence)

bullets = []
num_bullets = random.choices([0,1,2], weights=[0.4, 0.4, 0.2])[0]
for _ in range(num_bullets):
    action = random.choice(BULLET_ACTIONS)
    metric = random.choice(BULLET_METRICS)
    by = random.choice(["by 25%", "by 40%", "three-fold", "significantly", "measurably"])
    kw = random.choice(keywords_used + OTHER_PHRASES)
    bullets.append(f"• {action} {metric} {by} through the implementation of {kw}.")

text_parts = sentences + bullets
random.shuffle(text_parts)
text = " ".join(text_parts)
text = truncate_to_word_limit(text, max_words)
text = re.sub(r'\s+', ' ', text).strip()
return text, keywords_used

def generate_balanced_dataset(per_class: int = 100, enforce_unique_texts: bool = True, max_attempts_per_item: int = 50):
    records = []
    seen_texts: Set[str] = set()

    for job_cat in ALL_CATEGORIES:
        for positive in [True, False]:
            print(f"Generating {per_class} {'positive' if positive else 'negative'} samples for {job_cat}...")
            produced = 0
            attempts = 0
            while produced < per_class and attempts < per_class * max_attempts_per_item:
                attempts += 1
                name = fake.name()
                email = make_email(name)
                resume_text, kws_used = generate_resume_text(job_cat, positive)

                if enforce_unique_texts and resume_text in seen_texts:
                    continue

                text_lower = resume_text.lower()
                target_kw_present = any(re.search(r'\b' + re.escape(kw.lower()) + r'\b', text_lower) for kw in JOB_KEYWORDS)

                if positive and not target_kw_present:
                    continue
                if (not positive) and target_kw_present:
                    continue

                records.append({
                    "name": name,
                    "email": email,
                    "job_category": job_cat,
                    "resume_text": resume_text,
                    "keywords_used": kws_used,
                    "label": 1 if positive else 0
                })
                produced += 1
                if enforce_unique_texts:
                    seen_texts.add(resume_text)

            if produced < per_class:
                print(f"Warning: for category '{job_cat}' positive={positive} only produced {produced}/{per_class} after {attempts} attempts")

    df = pd.DataFrame(records)
    df = df.sample(frac=1, random_state=42).reset_index(drop=True)

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return df

# --- Main Execution ---
print("Generating enhanced balanced dataset...")
df = generate_balanced_dataset(per_class=1500)
print("\n" + "="*50)
print(f"Generated a total of {len(df)} records.")
print("\nLabel distribution:\n", df['label'].value_counts())
print("\nPer-category counts:\n", df['job_category'].value_counts())

df['word_count'] = df['resume_text'].apply(lambda t: len(t.split()))
print("\nMax words in resume_text:", df['word_count'].max())
print("\nSample record:\n", df.iloc[0].to_dict())

out_filename = "synthetic_resumes_enhanced.csv"
df.drop(columns=['word_count']).to_csv(out_filename, index=False)
print(f"\nSaved enhanced dataset to '{out_filename}'.")

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Generating enhanced balanced dataset...
Generating 1500 positive samples for Software Engineer...
Generating 1500 negative samples for Software Engineer...
Generating 1500 positive samples for Product Manager...
Generating 1500 negative samples for Product Manager...
Generating 1500 positive samples for Designer...
Generating 1500 negative samples for Designer...

=====
Generated a total of 9000 records.

Label distribution:
  label
0    4500
1    4500
Name: count, dtype: int64

Per-category counts:
  job_category
Designer      3000
Software Engineer  3000
Product Manager  3000
Name: count, dtype: int64

Max words in resume_text: 80

Sample record:
{'name': 'Hannah Newton', 'email': 'hannah.newton.9770@hotmail.com', 'job_category': 'Designer', 'resume_text': '• Pub'
Saved enhanced dataset to 'synthetic_resumes_enhanced.csv'.

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import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader, random_split
from torch.optim import AdamW
import pandas as pd
from sklearn.metrics import accuracy_score, f1_score
import warnings
import numpy as np
from tqdm.notebook import tqdm
import json
import copy
import math

!pip install opacus -q
from opacus.accountants import RDPAccountant

warnings.simplefilter("ignore")

# Load data
try:
    df = pd.read_csv("synthetic_resumes_enhanced.csv")
except FileNotFoundError:
    print("Error: 'synthetic_resumes_enhanced.csv' not found.")
    exit()

gpu_device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
cpu_device = torch.device("cpu")

job_categories = df['job_category'].unique().tolist()

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job2idx = {j: i for i, j in enumerate(job_categories)}
df['job_idx'] = df['job_category'].map(job2idx)

def tokenize(text): return text.lower().split()
all_tokens = [tok for text in df['resume_text'] for tok in tokenize(text)]
vocab = {tok: idx + 1 for idx, tok in enumerate(set(all_tokens))}
UNK_IDX = len(vocab) + 1
VOCAB_SIZE = UNK_IDX + 1
def encode(text): return [vocab.get(tok, UNK_IDX) for tok in tokenize(text)]

class ResumeDataset(Dataset):
    def __init__(self, texts, cat_indices, labels):
        self.texts = [torch.tensor(encode(t), dtype=torch.long) for t in texts]
        self.cats = torch.tensor(cat_indices, dtype=torch.long)
        self.labels = torch.tensor(labels, dtype=torch.float32)
    def __len__(self): return len(self.labels)
    def __getitem__(self, idx): return self.texts[idx], self.cats[idx], self.labels[idx]

def collate_bow(batch):
    texts, cats, labels = zip(*batch)
    labels = torch.tensor(labels, dtype=torch.float32)
    cats = torch.tensor(cats, dtype=torch.long)
    offsets = [0] + [len(text) for text in texts]
    offsets = torch.tensor(offsets[:-1]).cumsum(dim=0)
    texts = torch.cat(texts)
    return texts, offsets, cats, labels

dataset = ResumeDataset(df['resume_text'].tolist(), df['job_idx'].tolist(), df['label'].tolist())
train_size = int(0.7 * len(dataset))
val_size = int(0.15 * len(dataset))
test_size = len(dataset) - train_size - val_size
train_dataset, val_dataset, test_dataset = random_split(
    dataset, [train_size, val_size, test_size],
    generator=torch.Generator().manual_seed(42)
)

# BoW
class BagOfWordsClassifier(nn.Module):
    def __init__(self, vocab_size, num_categories, embed_dim=256, cat_embed_dim=64):
        super().__init__()
        self.text_embedding = nn.EmbeddingBag(vocab_size, embed_dim, sparse=False, padding_idx=0)
        self.cat_embedding = nn.Embedding(num_categories, cat_embed_dim)
        self.classifier = nn.Sequential(
            nn.Linear(embed_dim + cat_embed_dim, 256),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(256, 1)
        )
    def forward(self, texts, offsets, cat_idx):
        text_vec = self.text_embedding(texts, offsets)
        cat_vec = self.cat_embedding(cat_idx)
        combined = torch.cat([text_vec, cat_vec], dim=1)
        logits = self.classifier(combined)
        return logits.squeeze(-1)

# Helper Funcs
def eval_model(model, loader, device):
    model.eval()
    y_true, y_pred = [], []
    with torch.no_grad():
        for texts, offsets, cats, labels in loader:
            texts, offsets, cats, labels = texts.to(device), offsets.to(device), cats.to(device), labels.to(device)
            logits = model(texts, offsets, cats)
            y_hat = torch.sigmoid(logits).cpu().numpy()
            y_true.extend(labels.cpu().numpy())
            y_pred.extend((y_hat > 0.5).astype(int))
    return {"accuracy": accuracy_score(y_true, y_pred), "f1": f1_score(y_true, y_pred, zero_division=0)}

def train_baseline_robust(config):
    print("\n" + "="*50 + "\nRunning Improved Non-Private Baseline Training...\n" + "="*50)
    train_loader = DataLoader(train_dataset, batch_size=config['batch_size'], shuffle=True, collate_fn=collate_bow)
    val_loader = DataLoader(val_dataset, batch_size=config['batch_size'], collate_fn=collate_bow)
    test_loader = DataLoader(test_dataset, batch_size=config['batch_size'], collate_fn=collate_bow)
    model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(gpu_device)
    optimizer = AdamW(model.parameters(), lr=config['lr'])
    scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='max', factor=0.1, patience=3)
    criterion = nn.BCEWithLogitsLoss()
    best_val_f1 = 0
    for epoch in range(1, num_epochs + 1):
        # Training
        model.train()
        train_loader.sampler.set_epoch(epoch)
        train_loss = 0
        for texts, offsets, cats, labels in train_loader:
            texts, offsets, cats, labels = texts.to(device), offsets.to(device), cats.to(device), labels.to(device)
            logits = model(texts, offsets, cats)
            loss = criterion(logits, labels)
            train_loss += loss.item()
        train_loss /= len(train_loader)

        # Validation
        model.eval()
        val_loader.sampler.set_epoch(epoch)
        val_loss = 0
        val_f1 = 0
        for texts, offsets, cats, labels in val_loader:
            texts, offsets, cats, labels = texts.to(device), offsets.to(device), cats.to(device), labels.to(device)
            logits = model(texts, offsets, cats)
            val_loss += criterion(logits, labels).item()
            y_hat = torch.sigmoid(logits).cpu().numpy()
            y_true = labels.cpu().numpy()
            val_f1 += f1_score(y_true, y_hat, zero_division=0)
        val_loss /= len(val_loader)
        val_f1 /= len(val_loader)

        # Testing
        model.eval()
        test_loader.sampler.set_epoch(epoch)
        test_loss = 0
        test_f1 = 0
        for texts, offsets, cats, labels in test_loader:
            texts, offsets, cats, labels = texts.to(device), offsets.to(device), cats.to(device), labels.to(device)
            logits = model(texts, offsets, cats)
            test_loss += criterion(logits, labels).item()
            y_hat = torch.sigmoid(logits).cpu().numpy()
            y_true = labels.cpu().numpy()
            test_f1 += f1_score(y_true, y_hat, zero_division=0)
        test_loss /= len(test_loader)
        test_f1 /= len(test_loader)

        # Logging
        print(f"Epoch {epoch}: train_loss={train_loss:.4f}, val_loss={val_loss:.4f}, test_loss={test_loss:.4f}, val_f1={val_f1:.4f}, test_f1={test_f1:.4f}")

        # Early stopping
        if val_f1 > best_val_f1:
            best_val_f1 = val_f1
            best_model = model.clone()

    return best_model, best_val_f1, test_f1

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best_val_f1 = -1
patience_counter = 0
best_model_state = None
for epoch in range(config['epochs']):
    model.train()
    for texts, offsets, cats, labels in tqdm(train_loader, desc=f"Epoch {epoch+1}/{config['epochs']}"):
        texts, offsets, cats, labels = texts.to(gpu_device), offsets.to(gpu_device), cats.to(gpu_device), labels.to(
        optimizer.zero_grad()
        logits = model(texts, offsets, cats)
        loss = criterion(logits, labels)
        loss.backward()
        optimizer.step()
    val_metrics = eval_model(model, val_loader, gpu_device)
    val_f1 = val_metrics['f1']
    print(f"Epoch {epoch+1}: Val F1: {val_f1:.4f}, Current LR: {optimizer.param_groups[0]['lr']:.2E}")
    scheduler.step(val_f1)
    if val_f1 > best_val_f1:
        best_val_f1 = val_f1
        patience_counter = 0
        best_model_state = copy.deepcopy(model.state_dict())
    else:
        patience_counter += 1
        if patience_counter >= config['patience']:
            break
    model.load_state_dict(best_model_state)
    test_metrics = eval_model(model, test_loader, gpu_device)
    print(f"\nBASELINE RESULTS:\n -> Final Test Accuracy: {test_metrics['accuracy']:.4f}\n -> Final Test F1 Score: {te
    return test_metrics

def find_adaptive_clip_norm(model, loader, device):
    print("\n--- Running epoch to find adaptive clipping norm ---")
    model.train()
    criterion = nn.BCEWithLogitsLoss()
    norms = []
    for texts, offsets, cats, labels in tqdm(loader, desc="Analyzing Gradients"):
        batch_size = len(labels)
        for i in range(batch_size):
            text_end = offsets[i+1] if i+1 < len(offsets) else len(texts)
            text_i, offset_i, cat_i, label_i = texts[offsets[i]:text_end], torch.tensor([0]), cats[i:i+1], labels[i:i+1]
            model.zero_grad()
            logit_i = model(text_i.to(device), offset_i.to(device), cat_i.to(device))
            loss_i = criterion(logit_i, label_i.to(device))
            loss_i.backward()
            total_norm = sum(p.grad.norm(2).item() ** 2 for p in model.parameters() if p.grad is not None) ** 0.5
            norms.append(total_norm)
    median_norm = np.median(norms)
    print(f"--- Found median gradient norm: {median_norm:.4f} ---")
    return median_norm

def train_manual_dp(model, loader, optimizer, criterion, device, clip_norm, noise_multiplier):
    model.train()
    for texts, offsets, cats, labels in tqdm(loader, desc="DP Batch"):
        batch_size = len(labels)
        saved_grads = {name: [] for name, param in model.named_parameters() if param.requires_grad}
        for i in range(batch_size):
            optimizer.zero_grad()
            text_end = offsets[i+1] if i+1 < len(offsets) else len(texts)
            text_i, offset_i, cat_i, label_i = texts[offsets[i]:text_end], torch.tensor([0]), cats[i:i+1], labels[i:i+1]
            logit_i = model(text_i.to(device), offset_i.to(device), cat_i.to(device))
            loss_i = criterion(logit_i, label_i.to(device))
            loss_i.backward()
            for name, param in model.named_parameters():
                if param.requires_grad and param.grad is not None:
                    saved_grads[name].append(param.grad.clone())
        optimizer.zero_grad()
        for i in range(batch_size):
            total_norm = sum(saved_grads[name][i].norm(2).item() ** 2 for name, _ in model.named_parameters() if name in
            clip_factor = min(1.0, clip_norm / (total_norm + 1e-6))
            for name, _ in model.named_parameters():
                if name in saved_grads and i < len(saved_grads[name]):
                    saved_grads[name][i].mul_(clip_factor)
        for name, param in model.named_parameters():
            if name in saved_grads and saved_grads[name]:
                clipped_sum = torch.stack(saved_grads[name], dim=0).sum(dim=0)
                noise = torch.normal(0, clip_norm * noise_multiplier, size=clipped_sum.shape, device=device)
                param.grad = (clipped_sum + noise) / batch_size
        optimizer.step()

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if __name__ == '__main__':

    baseline_config = {'lr': 1e-3, 'batch_size': 128, 'epochs': 20, 'patience': 5}
    baseline_metrics = train_baseline_robust(baseline_config)

    DP_BATCH_SIZE = 256
    DP_EPOCHS = 8

    temp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
    temp_loader = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
    adaptive_c = find_adaptive_clip_norm(temp_model, temp_loader, cpu_device)

    # Custom SWEEP grid
    CLIP_NORMS = sorted(list(set([0.5, 1.0, round(adaptive_c, 2)])))
    NOISE_MULTIPLIERS = [0.8, 1.2, 1.5]
    LEARNING_RATES = [5e-3, 1e-4]

    print(f"\nStarting grid sweep with C values: {CLIP_NORMS}")

    results = []

    for lr in LEARNING_RATES:
        for c in CLIP_NORMS:
            for sigma in NOISE_MULTIPLIERS:
                print("\n" + "-" * 50 + f"\nStarting DP-SGD Sweep Run...")
                print(f"Params: C={c},  $\sigma$ ={sigma}, BS={DP_BATCH_SIZE}, LR={lr}")
                print("-" * 50)

                dp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
                dp_optimizer = AdamW(dp_model.parameters(), lr=lr)
                dp_criterion = nn.BCEWithLogitsLoss()

                train_loader_cpu = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
                test_loader_cpu = DataLoader(test_dataset, batch_size=DP_BATCH_SIZE, collate_fn=collate_bow)

                accountant = RDPAccountant()
                DELTA = 1 / len(train_dataset)
                sample_rate = DP_BATCH_SIZE / len(train_dataset)

                for epoch in range(DP_EPOCHS):
                    train_manual_dp(dp_model, train_loader_cpu, dp_optimizer, dp_criterion, cpu_device, c, sigma)
                    accountant.step(noise_multiplier=sigma, sample_rate=sample_rate)
                    epsilon = accountant.get_epsilon(delta=DELTA)
                    print(f"Epoch {epoch+1}/{DP_EPOCHS} | Epsilon: {epsilon:.2f}")

                dp_metrics = eval_model(dp_model, test_loader_cpu, cpu_device)
                final_epsilon = accountant.get_epsilon(delta=DELTA)

                print("\n" + "-" * 15)
                print(f"Result for Run (C={c},  $\sigma$ ={sigma}, LR={lr}):")
                print(f"  -> Final Test Accuracy: {dp_metrics['accuracy']:.4f}")
                print(f"  -> Final Test F1 Score: {dp_metrics['f1']:.4f}")
                print(f"  -> Final Epsilon ( $\epsilon$ ): {final_epsilon:.2f}")
                print("-" * 15)

                results.append({
                    "lr": lr, "clip_norm": c, "noise_multiplier": sigma,
                    "final_epsilon": final_epsilon, "accuracy": dp_metrics['accuracy'], "f1": dp_metrics['f1']
                })

    # 3. Report Final Comparison
    results_df = pd.DataFrame(results).rename(columns={"accuracy": "test_accuracy", "f1": "test_f1"})
    display_cols = ['lr', 'clip_norm', 'noise_multiplier', 'final_epsilon', 'test_accuracy', 'test_f1']
    results_df = results_df[display_cols].sort_values(by="test_f1", ascending=False)

    print("\n\n" + "=" * 80 + "\n          Final Tuned Sweep Results: Baseline vs. DP-SGD\n" + "=" * 80)
    print(f"BASELINE MODEL (Bag-of-Words, Non-Private):")
    print(f"  Accuracy: {baseline_metrics['accuracy']:.4f}")
    print(f"  F1 Score: {baseline_metrics['f1']:.4f}\n")
    print(f"DP-SGD MODELS (Bag-of-Words, Private,  $\delta \approx$  {DELTA:.2E}):")
    print(results_df.to_string(index=False))
    print("=" * 80)

```





Using device for Baseline: cuda  
Using device for DP-SGD: cpu

=====

Running Improved Non-Private Baseline Training...

=====

|                                                |                                 |
|------------------------------------------------|---------------------------------|
| Epoch 1/50: 100%                               | 50/50 [00:00<00:00, 94.57it/s]  |
| Epoch 1: Val F1: 0.8058, Current LR: 1.00E-03  |                                 |
| Epoch 2/50: 100%                               | 50/50 [00:00<00:00, 72.69it/s]  |
| Epoch 2: Val F1: 0.9434, Current LR: 1.00E-03  |                                 |
| Epoch 3/50: 100%                               | 50/50 [00:00<00:00, 100.78it/s] |
| Epoch 3: Val F1: 0.9843, Current LR: 1.00E-03  |                                 |
| Epoch 4/50: 100%                               | 50/50 [00:00<00:00, 102.51it/s] |
| Epoch 4: Val F1: 0.9943, Current LR: 1.00E-03  |                                 |
| Epoch 5/50: 100%                               | 50/50 [00:00<00:00, 159.54it/s] |
| Epoch 5: Val F1: 0.9986, Current LR: 1.00E-03  |                                 |
| Epoch 6/50: 100%                               | 50/50 [00:00<00:00, 147.40it/s] |
| Epoch 6: Val F1: 0.9993, Current LR: 1.00E-03  |                                 |
| Epoch 7/50: 100%                               | 50/50 [00:00<00:00, 162.77it/s] |
| Epoch 7: Val F1: 1.0000, Current LR: 1.00E-03  |                                 |
| Epoch 8/50: 100%                               | 50/50 [00:00<00:00, 164.24it/s] |
| Epoch 8: Val F1: 1.0000, Current LR: 1.00E-03  |                                 |
| Epoch 9/50: 100%                               | 50/50 [00:00<00:00, 152.30it/s] |
| Epoch 9: Val F1: 1.0000, Current LR: 1.00E-03  |                                 |
| Epoch 10/50: 100%                              | 50/50 [00:00<00:00, 158.51it/s] |
| Epoch 10: Val F1: 1.0000, Current LR: 1.00E-03 |                                 |
| Epoch 11/50: 100%                              | 50/50 [00:00<00:00, 166.21it/s] |
| Epoch 11: Val F1: 1.0000, Current LR: 1.00E-03 |                                 |
| Epoch 12/50: 100%                              | 50/50 [00:00<00:00, 164.19it/s] |
| Epoch 12: Val F1: 1.0000, Current LR: 1.00E-04 |                                 |
| --- Early stopping triggered. ---              |                                 |

Loading best model and evaluating on the test set...

IMPROVED BASELINE RESULTS:

-> Final Test Accuracy: 0.9993  
-> Final Test F1 Score: 0.9993

--- Running epoch to find adaptive clipping norm ---

|                           |                               |
|---------------------------|-------------------------------|
| Analyzing Gradients: 100% | 25/25 [00:13<00:00, 2.34it/s] |
|---------------------------|-------------------------------|

--- Found median gradient norm: 3.6721 ---

Starting grid sweep with C values: [0.5, 1.0, np.float64(3.67)]

-----

Starting DP-SGD Sweep Run...

Params: C=0.5,  $\sigma$ =0.8, BS=256, LR=0.005

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:22<00:00, 1.87it/s] |
| Epoch 1/8   Epsilon: 1.73 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.02it/s] |
| Epoch 2/8   Epsilon: 1.87 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 3/8   Epsilon: 1.98 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 4/8   Epsilon: 2.06 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.21it/s] |
| Epoch 5/8   Epsilon: 2.13 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.22it/s] |
| Epoch 6/8   Epsilon: 2.19 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.26it/s] |
| Epoch 7/8   Epsilon: 2.24 |                               |
| Manual DP Batch: 100%     | 25/25 [00:16<00:00, 1.93it/s] |
| Epoch 8/8   Epsilon: 2.30 |                               |

-----

Result for Run (C=0.5,  $\sigma$ =0.8, LR=0.005):

```
-----  
-> Final Test Accuracy: 0.7548  
-> Final Test F1 Score: 0.7005  
-> Final Epsilon ( $\epsilon$ ): 2.30  
-----
```

```
-----  
Starting DP-SGD Sweep Run...  
Params: C=0.5,  $\sigma$ =1.2, BS=256, LR=0.005  
-----
```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.75it/s] |
| Epoch 1/8   Epsilon: 0.68 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.86it/s] |
| Epoch 2/8   Epsilon: 0.72 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.02it/s] |
| Epoch 3/8   Epsilon: 0.75 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 4/8   Epsilon: 0.78 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 5/8   Epsilon: 0.80 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 6/8   Epsilon: 0.82 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 7/8   Epsilon: 0.84 |                               |
| Manual DP Batch: 100%     | 25/25 [00:17<00:00, 1.72it/s] |
| Epoch 8/8   Epsilon: 0.86 |                               |

```
-----  
Result for Run (C=0.5,  $\sigma$ =1.2, LR=0.005):  
-> Final Test Accuracy: 0.5059  
-> Final Test F1 Score: 0.4103  
-> Final Epsilon ( $\epsilon$ ): 0.86  
-----
```

```
-----  
Starting DP-SGD Sweep Run...  
Params: C=0.5,  $\sigma$ =1.5, BS=256, LR=0.005  
-----
```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.94it/s] |
| Epoch 1/8   Epsilon: 0.40 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.93it/s] |
| Epoch 2/8   Epsilon: 0.42 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.03it/s] |
| Epoch 3/8   Epsilon: 0.44 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.08it/s] |
| Epoch 4/8   Epsilon: 0.45 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 5/8   Epsilon: 0.47 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 6/8   Epsilon: 0.48 |                               |
| Manual DP Batch: 100%     | 25/25 [00:16<00:00, 1.29it/s] |
| Epoch 7/8   Epsilon: 0.49 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.88it/s] |
| Epoch 8/8   Epsilon: 0.51 |                               |

```
-----  
Result for Run (C=0.5,  $\sigma$ =1.5, LR=0.005):  
-> Final Test Accuracy: 0.5000  
-> Final Test F1 Score: 0.4063  
-> Final Epsilon ( $\epsilon$ ): 0.51  
-----
```

```
-----  
Starting DP-SGD Sweep Run...  
Params: C=1.0,  $\sigma$ =0.8, BS=256, LR=0.005  
-----
```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.98it/s] |
| Epoch 1/8   Epsilon: 1.73 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.07it/s] |
| Epoch 2/8   Epsilon: 1.87 |                               |

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 3/8   Epsilon: 1.98 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.78it/s] |
| Epoch 4/8   Epsilon: 2.06 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 5/8   Epsilon: 2.13 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 6/8   Epsilon: 2.19 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.86it/s] |
| Epoch 7/8   Epsilon: 2.24 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.79it/s] |
| Epoch 8/8   Epsilon: 2.30 |                               |

-----  
Result for Run (C=1.0,  $\sigma$ =0.8, LR=0.005):  
-> Final Test Accuracy: 0.6393  
-> Final Test F1 Score: 0.5086  
-> Final Epsilon ( $\epsilon$ ): 2.30  
-----

-----  
Starting DP-SGD Sweep Run...  
Params: C=1.0,  $\sigma$ =1.2, BS=256, LR=0.005  
-----

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.97it/s] |
| Epoch 1/8   Epsilon: 0.68 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.04it/s] |
| Epoch 2/8   Epsilon: 0.72 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 3/8   Epsilon: 0.75 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 4/8   Epsilon: 0.78 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 5/8   Epsilon: 0.80 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 6/8   Epsilon: 0.82 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.92it/s] |
| Epoch 7/8   Epsilon: 0.84 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.69it/s] |
| Epoch 8/8   Epsilon: 0.86 |                               |

-----  
Result for Run (C=1.0,  $\sigma$ =1.2, LR=0.005):  
-> Final Test Accuracy: 0.5007  
-> Final Test F1 Score: 0.4077  
-> Final Epsilon ( $\epsilon$ ): 0.86  
-----

-----  
Starting DP-SGD Sweep Run...  
Params: C=1.0,  $\sigma$ =1.5, BS=256, LR=0.005  
-----

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.94it/s] |
| Epoch 1/8   Epsilon: 0.40 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.03it/s] |
| Epoch 2/8   Epsilon: 0.42 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.04it/s] |
| Epoch 3/8   Epsilon: 0.44 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.07it/s] |
| Epoch 4/8   Epsilon: 0.45 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 5/8   Epsilon: 0.47 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 6/8   Epsilon: 0.48 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 7/8   Epsilon: 0.49 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.97it/s] |

Epoch 8/8 | Epsilon: 0.51

Result for Run (C=1.0,  $\sigma$ =1.5, LR=0.005):

-> Final Test Accuracy: 0.5541  
-> Final Test F1 Score: 0.4537  
-> Final Epsilon ( $\epsilon$ ): 0.51

Starting DP-SGD Sweep Run...

Params: C=3.67,  $\sigma$ =0.8, BS=256, LR=0.005

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.57it/s] |
| Epoch 1/8   Epsilon: 1.73 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.83it/s] |
| Epoch 2/8   Epsilon: 1.87 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.95it/s] |
| Epoch 3/8   Epsilon: 1.98 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.05it/s] |
| Epoch 4/8   Epsilon: 2.06 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.06it/s] |
| Epoch 5/8   Epsilon: 2.13 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 6/8   Epsilon: 2.19 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 7/8   Epsilon: 2.24 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.05it/s] |
| Epoch 8/8   Epsilon: 2.30 |                               |

Result for Run (C=3.67,  $\sigma$ =0.8, LR=0.005):

-> Final Test Accuracy: 0.8222  
-> Final Test F1 Score: 0.8187  
-> Final Epsilon ( $\epsilon$ ): 2.30

Starting DP-SGD Sweep Run...

Params: C=3.67,  $\sigma$ =1.2, BS=256, LR=0.005

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.87it/s] |
| Epoch 1/8   Epsilon: 0.68 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.77it/s] |
| Epoch 2/8   Epsilon: 0.72 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 1.96it/s] |
| Epoch 3/8   Epsilon: 0.75 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.07it/s] |
| Epoch 4/8   Epsilon: 0.78 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.06it/s] |
| Epoch 5/8   Epsilon: 0.80 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 6/8   Epsilon: 0.82 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 7/8   Epsilon: 0.84 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.07it/s] |
| Epoch 8/8   Epsilon: 0.86 |                               |

Result for Run (C=3.67,  $\sigma$ =1.2, LR=0.005):

-> Final Test Accuracy: 0.6919  
-> Final Test F1 Score: 0.7075  
-> Final Epsilon ( $\epsilon$ ): 0.86

Starting DP-SGD Sweep Run...

Params: C=3.67,  $\sigma$ =1.5, BS=256, LR=0.005

|                       |                               |
|-----------------------|-------------------------------|
| Manual DP Batch: 100% | 25/25 [00:14<00:00, 1.84it/s] |
|-----------------------|-------------------------------|

Epoch 1/8 | Epsilon: 0.40  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.80it/s]  
Epoch 2/8 | Epsilon: 0.42  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.96it/s]  
Epoch 3/8 | Epsilon: 0.44  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.04it/s]  
Epoch 4/8 | Epsilon: 0.45  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.07it/s]  
Epoch 5/8 | Epsilon: 0.47  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.82it/s]  
Epoch 6/8 | Epsilon: 0.48  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.08it/s]  
Epoch 7/8 | Epsilon: 0.49  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]  
Epoch 8/8 | Epsilon: 0.51

-----  
Result for Run (C=3.67,  $\sigma$ =1.5, LR=0.005):  
-> Final Test Accuracy: 0.7111  
-> Final Test F1 Score: 0.6991  
-> Final Epsilon ( $\epsilon$ ): 0.51  
-----

-----  
Starting DP-SGD Sweep Run...  
Params: C=0.5,  $\sigma$ =0.8, BS=256, LR=0.0001  
-----

Manual DP Batch: 100% 25/25 [00:14<00:00, 1.78it/s]  
Epoch 1/8 | Epsilon: 1.73  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.84it/s]  
Epoch 2/8 | Epsilon: 1.87  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.01it/s]  
Epoch 3/8 | Epsilon: 1.98  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.07it/s]  
Epoch 4/8 | Epsilon: 2.06  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 5/8 | Epsilon: 2.13  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 6/8 | Epsilon: 2.19  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]  
Epoch 7/8 | Epsilon: 2.24  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]  
Epoch 8/8 | Epsilon: 2.30

-----  
Result for Run (C=0.5,  $\sigma$ =0.8, LR=0.0001):  
-> Final Test Accuracy: 0.4919  
-> Final Test F1 Score: 0.5842  
-> Final Epsilon ( $\epsilon$ ): 2.30  
-----

-----  
Starting DP-SGD Sweep Run...  
Params: C=0.5,  $\sigma$ =1.2, BS=256, LR=0.0001  
-----

Manual DP Batch: 100% 25/25 [00:14<00:00, 1.83it/s]  
Epoch 1/8 | Epsilon: 0.68  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.81it/s]  
Epoch 2/8 | Epsilon: 0.72  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.93it/s]  
Epoch 3/8 | Epsilon: 0.75  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.03it/s]  
Epoch 4/8 | Epsilon: 0.78  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 5/8 | Epsilon: 0.80  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 6/8 | Epsilon: 0.82

```
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.08it/s]
Epoch 7/8 | Epsilon: 0.84
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.10it/s]
Epoch 8/8 | Epsilon: 0.86
```

```
-----
Result for Run (C=0.5,  $\sigma$ =1.2, LR=0.0001):
-> Final Test Accuracy: 0.5185
-> Final Test F1 Score: 0.5855
-> Final Epsilon ( $\epsilon$ ): 0.86
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=0.5,  $\sigma$ =1.5, BS=256, LR=0.0001
-----
```

```
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.00it/s]
Epoch 1/8 | Epsilon: 0.40
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.78it/s]
Epoch 2/8 | Epsilon: 0.42
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.92it/s]
Epoch 3/8 | Epsilon: 0.44
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.00it/s]
Epoch 4/8 | Epsilon: 0.45
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.05it/s]
Epoch 5/8 | Epsilon: 0.47
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.11it/s]
Epoch 6/8 | Epsilon: 0.48
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.10it/s]
Epoch 7/8 | Epsilon: 0.49
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.13it/s]
Epoch 8/8 | Epsilon: 0.51
```

```
-----
Result for Run (C=0.5,  $\sigma$ =1.5, LR=0.0001):
-> Final Test Accuracy: 0.5244
-> Final Test F1 Score: 0.3462
-> Final Epsilon ( $\epsilon$ ): 0.51
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=1.0,  $\sigma$ =0.8, BS=256, LR=0.0001
-----
```

```
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.11it/s]
Epoch 1/8 | Epsilon: 1.73
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.92it/s]
Epoch 2/8 | Epsilon: 1.87
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.69it/s]
Epoch 3/8 | Epsilon: 1.98
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.92it/s]
Epoch 4/8 | Epsilon: 2.06
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.74it/s]
Epoch 5/8 | Epsilon: 2.13
Manual DP Batch: 100%                25/25 [00:19<00:00, 2.09it/s]
Epoch 6/8 | Epsilon: 2.19
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.92it/s]
Epoch 7/8 | Epsilon: 2.24
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.71it/s]
Epoch 8/8 | Epsilon: 2.30
```

```
-----
Result for Run (C=1.0,  $\sigma$ =0.8, LR=0.0001):
-> Final Test Accuracy: 0.4978
-> Final Test F1 Score: 0.1567
-> Final Epsilon ( $\epsilon$ ): 2.30
-----
```

```
-----
Starting DP-SGD Sweep Run...
```

```
-----
Params: C=1.0,  $\sigma$ =1.2, BS=256, LR=0.0001
-----

Manual DP Batch: 100%                                25/25 [00:13<00:00, 1.96it/s]
Epoch 1/8 | Epsilon: 0.68
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.07it/s]
Epoch 2/8 | Epsilon: 0.72
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 3/8 | Epsilon: 0.75
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.09it/s]
Epoch 4/8 | Epsilon: 0.78
Manual DP Batch: 100%                                25/25 [00:14<00:00, 2.09it/s]
Epoch 5/8 | Epsilon: 0.80
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.14it/s]
Epoch 6/8 | Epsilon: 0.82
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.78it/s]
Epoch 7/8 | Epsilon: 0.84
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.86it/s]
Epoch 8/8 | Epsilon: 0.86

-----
Result for Run (C=1.0,  $\sigma$ =1.2, LR=0.0001):
-> Final Test Accuracy: 0.5111
-> Final Test F1 Score: 0.1582
-> Final Epsilon ( $\epsilon$ ): 0.86
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=1.0,  $\sigma$ =1.5, BS=256, LR=0.0001
-----

Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.76it/s]
Epoch 1/8 | Epsilon: 0.40
Manual DP Batch: 100%                                25/25 [00:13<00:00, 1.94it/s]
Epoch 2/8 | Epsilon: 0.42
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.03it/s]
Epoch 3/8 | Epsilon: 0.44
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.07it/s]
Epoch 4/8 | Epsilon: 0.45
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 5/8 | Epsilon: 0.47
Manual DP Batch: 100%                                25/25 [00:14<00:00, 2.11it/s]
Epoch 6/8 | Epsilon: 0.48
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.12it/s]
Epoch 7/8 | Epsilon: 0.49
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.13it/s]
Epoch 8/8 | Epsilon: 0.51

-----
Result for Run (C=1.0,  $\sigma$ =1.5, LR=0.0001):
-> Final Test Accuracy: 0.5178
-> Final Test F1 Score: 0.3899
-> Final Epsilon ( $\epsilon$ ): 0.51
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=3.67,  $\sigma$ =0.8, BS=256, LR=0.0001
-----

Manual DP Batch: 100%                                25/25 [00:16<00:00, 1.71it/s]
Epoch 1/8 | Epsilon: 1.73
Manual DP Batch: 100%                                25/25 [00:14<00:00, 2.02it/s]
Epoch 2/8 | Epsilon: 1.87
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.06it/s]
Epoch 3/8 | Epsilon: 1.98
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 4/8 | Epsilon: 2.06
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.12it/s]
```

```
Epoch 5/8 | Epsilon: 2.13
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.11it/s]
Epoch 6/8 | Epsilon: 2.19
Manual DP Batch: 100%                25/25 [00:15<00:00, 1.21it/s]
Epoch 7/8 | Epsilon: 2.24
Manual DP Batch: 100%                25/25 [00:24<00:00, 1.40it/s]
Epoch 8/8 | Epsilon: 2.30
```

```
-----
Result for Run (C=3.67,  $\sigma$ =0.8, LR=0.0001):
-> Final Test Accuracy: 0.5526
-> Final Test F1 Score: 0.4248
-> Final Epsilon ( $\epsilon$ ): 2.30
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=3.67,  $\sigma$ =1.2, BS=256, LR=0.0001
-----
```

```
Manual DP Batch: 100%                25/25 [00:26<00:00, 1.12it/s]
Epoch 1/8 | Epsilon: 0.68
Manual DP Batch: 100%                25/25 [00:18<00:00, 2.00it/s]
Epoch 2/8 | Epsilon: 0.72
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.96it/s]
Epoch 3/8 | Epsilon: 0.75
Manual DP Batch: 100%                25/25 [00:24<00:00, 1.22s/it]
Epoch 4/8 | Epsilon: 0.78
Manual DP Batch: 100%                25/25 [00:20<00:00, 1.85it/s]
Epoch 5/8 | Epsilon: 0.80
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.13it/s]
Epoch 6/8 | Epsilon: 0.82
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.11it/s]
Epoch 7/8 | Epsilon: 0.84
Manual DP Batch: 100%                25/25 [00:17<00:00, 1.96it/s]
Epoch 8/8 | Epsilon: 0.86
```

```
-----
Result for Run (C=3.67,  $\sigma$ =1.2, LR=0.0001):
-> Final Test Accuracy: 0.5007
-> Final Test F1 Score: 0.4200
-> Final Epsilon ( $\epsilon$ ): 0.86
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=3.67,  $\sigma$ =1.5, BS=256, LR=0.0001
-----
```

```
Manual DP Batch: 100%                25/25 [00:17<00:00, 1.09s/it]
Epoch 1/8 | Epsilon: 0.40
Manual DP Batch: 100%                25/25 [00:15<00:00, 2.09it/s]
Epoch 2/8 | Epsilon: 0.42
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.06it/s]
Epoch 3/8 | Epsilon: 0.44
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.75it/s]
Epoch 4/8 | Epsilon: 0.45
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.84it/s]
Epoch 5/8 | Epsilon: 0.47
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.00it/s]
Epoch 6/8 | Epsilon: 0.48
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.03it/s]
Epoch 7/8 | Epsilon: 0.49
Manual DP Batch: 100%                25/25 [00:14<00:00, 2.08it/s]
Epoch 8/8 | Epsilon: 0.51
```

```
-----
Result for Run (C=3.67,  $\sigma$ =1.5, LR=0.0001):
-> Final Test Accuracy: 0.5289
-> Final Test F1 Score: 0.5777
-> Final Epsilon ( $\epsilon$ ): 0.51
-----
```



```
-----
Starting DP-SGD Sweep Run...
Params: C=0.5,  $\sigma$ =0.8, BS=256, LR=5e-05
-----

Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 1/8 | Epsilon: 1.73
Manual DP Batch: 100%                                25/25 [00:14<00:00, 2.08it/s]
Epoch 2/8 | Epsilon: 1.87
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 3/8 | Epsilon: 1.98
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.80it/s]
Epoch 4/8 | Epsilon: 2.06
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.91it/s]
Epoch 5/8 | Epsilon: 2.13
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.00it/s]
Epoch 6/8 | Epsilon: 2.19
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.07it/s]
Epoch 7/8 | Epsilon: 2.24
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.08it/s]
Epoch 8/8 | Epsilon: 2.30

-----
Result for Run (C=0.5,  $\sigma$ =0.8, LR=5e-05):
-> Final Test Accuracy: 0.5067
-> Final Test F1 Score: 0.4459
-> Final Epsilon ( $\epsilon$ ): 2.30
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=0.5,  $\sigma$ =1.2, BS=256, LR=5e-05
-----

Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 1/8 | Epsilon: 0.68
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.09it/s]
Epoch 2/8 | Epsilon: 0.72
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.11it/s]
Epoch 3/8 | Epsilon: 0.75
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.87it/s]
Epoch 4/8 | Epsilon: 0.78
Manual DP Batch: 100%                                25/25 [00:14<00:00, 1.79it/s]
Epoch 5/8 | Epsilon: 0.80
Manual DP Batch: 100%                                25/25 [00:13<00:00, 1.94it/s]
Epoch 6/8 | Epsilon: 0.82
Manual DP Batch: 100%                                25/25 [00:14<00:00, 2.08it/s]
Epoch 7/8 | Epsilon: 0.84
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.08it/s]
Epoch 8/8 | Epsilon: 0.86

-----
Result for Run (C=0.5,  $\sigma$ =1.2, LR=5e-05):
-> Final Test Accuracy: 0.4993
-> Final Test F1 Score: 0.3550
-> Final Epsilon ( $\epsilon$ ): 0.86
-----
```

```
-----
Starting DP-SGD Sweep Run...
Params: C=0.5,  $\sigma$ =1.5, BS=256, LR=5e-05
-----

Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.11it/s]
Epoch 1/8 | Epsilon: 0.40
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.11it/s]
Epoch 2/8 | Epsilon: 0.42
Manual DP Batch: 100%                                25/25 [00:13<00:00, 2.10it/s]
Epoch 3/8 | Epsilon: 0.44
```

Manual DP Batch: 100%25/25 [00:14<00:00, 1.95it/s]

Epoch 4/8 | Epsilon: 0.45

Manual DP Batch: 100%25/25 [00:14<00:00, 1.72it/s]

Epoch 5/8 | Epsilon: 0.47

Manual DP Batch: 100%25/25 [00:14<00:00, 1.85it/s]

Epoch 6/8 | Epsilon: 0.48

Manual DP Batch: 100%25/25 [00:13<00:00, 2.01it/s]

Epoch 7/8 | Epsilon: 0.49

Manual DP Batch: 100%25/25 [00:14<00:00, 2.04it/s]

Epoch 8/8 | Epsilon: 0.51

-----

Result for Run (C=0.5,  $\sigma$ =1.5, LR=5e-05):

-> Final Test Accuracy: 0.4963

-> Final Test F1 Score: 0.1667

-> Final Epsilon ( $\epsilon$ ): 0.51

-----

-----

Starting DP-SGD Sweep Run...

Params: C=1.0,  $\sigma$ =0.8, BS=256, LR=5e-05

-----

Manual DP Batch: 100%25/25 [00:14<00:00, 1.94it/s]

Epoch 1/8 | Epsilon: 1.73

Manual DP Batch: 100%25/25 [00:14<00:00, 2.08it/s]

Epoch 2/8 | Epsilon: 1.87

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 3/8 | Epsilon: 1.98

Manual DP Batch: 100%25/25 [00:13<00:00, 1.96it/s]

Epoch 4/8 | Epsilon: 2.06

Manual DP Batch: 100%25/25 [00:14<00:00, 1.70it/s]

Epoch 5/8 | Epsilon: 2.13

Manual DP Batch: 100%25/25 [00:13<00:00, 1.90it/s]

Epoch 6/8 | Epsilon: 2.19

Manual DP Batch: 100%25/25 [00:14<00:00, 2.02it/s]

Epoch 7/8 | Epsilon: 2.24

Manual DP Batch: 100%25/25 [00:13<00:00, 2.07it/s]

Epoch 8/8 | Epsilon: 2.30

-----

Result for Run (C=1.0,  $\sigma$ =0.8, LR=5e-05):

-> Final Test Accuracy: 0.4985

-> Final Test F1 Score: 0.0000

-> Final Epsilon ( $\epsilon$ ): 2.30

-----

-----

Starting DP-SGD Sweep Run...

Params: C=1.0,  $\sigma$ =1.2, BS=256, LR=5e-05

-----

Manual DP Batch: 100%25/25 [00:13<00:00, 2.08it/s]

Epoch 1/8 | Epsilon: 0.68

Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 2/8 | Epsilon: 0.72

Manual DP Batch: 100%25/25 [00:14<00:00, 1.80it/s]

Epoch 3/8 | Epsilon: 0.75

Manual DP Batch: 100%25/25 [00:14<00:00, 1.92it/s]

Epoch 4/8 | Epsilon: 0.78

Manual DP Batch: 100%25/25 [00:14<00:00, 1.70it/s]

Epoch 5/8 | Epsilon: 0.80

Manual DP Batch: 100%25/25 [00:13<00:00, 1.90it/s]

Epoch 6/8 | Epsilon: 0.82

Manual DP Batch: 100%25/25 [00:13<00:00, 2.01it/s]

Epoch 7/8 | Epsilon: 0.84

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 8/8 | Epsilon: 0.86

```

Result for Run (C=1.0,  $\sigma$ =1.2, LR=5e-05):
-> Final Test Accuracy: 0.4963
-> Final Test F1 Score: 0.0000
-> Final Epsilon ( $\epsilon$ ): 0.86

```

```

Starting DP-SGD Sweep Run...
Params: C=1.0,  $\sigma$ =1.5, BS=256, LR=5e-05

```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.05it/s] |
| Epoch 1/8   Epsilon: 0.40 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 2/8   Epsilon: 0.42 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 3/8   Epsilon: 0.44 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 4/8   Epsilon: 0.45 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.79it/s] |
| Epoch 5/8   Epsilon: 0.47 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.88it/s] |
| Epoch 6/8   Epsilon: 0.48 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.00it/s] |
| Epoch 7/8   Epsilon: 0.49 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.06it/s] |
| Epoch 8/8   Epsilon: 0.51 |                               |

```

Result for Run (C=1.0,  $\sigma$ =1.5, LR=5e-05):
-> Final Test Accuracy: 0.5230
-> Final Test F1 Score: 0.5484
-> Final Epsilon ( $\epsilon$ ): 0.51

```

```

Starting DP-SGD Sweep Run...
Params: C=3.67,  $\sigma$ =0.8, BS=256, LR=5e-05

```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 1/8   Epsilon: 1.73 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.06it/s] |
| Epoch 2/8   Epsilon: 1.87 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.06it/s] |
| Epoch 3/8   Epsilon: 1.98 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.04it/s] |
| Epoch 4/8   Epsilon: 2.06 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.77it/s] |
| Epoch 5/8   Epsilon: 2.13 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 1.87it/s] |
| Epoch 6/8   Epsilon: 2.19 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.01it/s] |
| Epoch 7/8   Epsilon: 2.24 |                               |
| Manual DP Batch: 100%     | 25/25 [00:14<00:00, 2.09it/s] |
| Epoch 8/8   Epsilon: 2.30 |                               |

```

Result for Run (C=3.67,  $\sigma$ =0.8, LR=5e-05):
-> Final Test Accuracy: 0.5348
-> Final Test F1 Score: 0.5177
-> Final Epsilon ( $\epsilon$ ): 2.30

```

```

Starting DP-SGD Sweep Run...
Params: C=3.67,  $\sigma$ =1.2, BS=256, LR=5e-05

```

|                           |                               |
|---------------------------|-------------------------------|
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 1/8   Epsilon: 0.68 |                               |
| Manual DP Batch: 100%     | 25/25 [00:13<00:00, 2.07it/s] |

Epoch 2/8 | Epsilon: 0.72

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 3/8 | Epsilon: 0.75

Manual DP Batch: 100%25/25 [00:14<00:00, 1.91it/s]

Epoch 4/8 | Epsilon: 0.78

Manual DP Batch: 100%25/25 [00:14<00:00, 1.71it/s]

Epoch 5/8 | Epsilon: 0.80

Manual DP Batch: 100%25/25 [00:13<00:00, 1.91it/s]

Epoch 6/8 | Epsilon: 0.82

Manual DP Batch: 100%25/25 [00:14<00:00, 2.03it/s]

Epoch 7/8 | Epsilon: 0.84

Manual DP Batch: 100%25/25 [00:14<00:00, 2.06it/s]

Epoch 8/8 | Epsilon: 0.86

-----

Result for Run (C=3.67,  $\sigma$ =1.2, LR=5e-05):

-> Final Test Accuracy: 0.5252

-> Final Test F1 Score: 0.4479

-> Final Epsilon ( $\epsilon$ ): 0.86

-----

-----

Starting DP-SGD Sweep Run...

Params: C=3.67,  $\sigma$ =1.5, BS=256, LR=5e-05

-----

Manual DP Batch: 100%25/25 [00:14<00:00, 2.07it/s]

Epoch 1/8 | Epsilon: 0.40

Manual DP Batch: 100%25/25 [00:14<00:00, 2.05it/s]

Epoch 2/8 | Epsilon: 0.42

Manual DP Batch: 100%25/25 [00:14<00:00, 2.07it/s]

Epoch 3/8 | Epsilon: 0.44

Manual DP Batch: 100%25/25 [00:14<00:00, 1.77it/s]

Epoch 4/8 | Epsilon: 0.45

Manual DP Batch: 100%25/25 [00:14<00:00, 1.79it/s]

Epoch 5/8 | Epsilon: 0.47

Manual DP Batch: 100%25/25 [00:14<00:00, 1.96it/s]

Epoch 6/8 | Epsilon: 0.48

Manual DP Batch: 100%25/25 [00:14<00:00, 2.00it/s]

Epoch 7/8 | Epsilon: 0.49

Manual DP Batch: 100%25/25 [00:14<00:00, 2.06it/s]

Epoch 8/8 | Epsilon: 0.51

-----

Result for Run (C=3.67,  $\sigma$ =1.5, LR=5e-05):

-> Final Test Accuracy: 0.5096

-> Final Test F1 Score: 0.3119

-> Final Epsilon ( $\epsilon$ ): 0.51

-----

=====

Final Tuned Sweep Results: Baseline vs. DP-SGD

=====

BASELINE MODEL (Bag-of-Words, Non-Private):

Accuracy: 0.9993

F1 Score: 0.9993

DP-SGD MODELS (Bag-of-Words, Private,  $\delta$ =1.59E-04):

| lr      | clip_norm | noise_multiplier | final_epsilon | test_accuracy | test_f1  |
|---------|-----------|------------------|---------------|---------------|----------|
| 0.00500 | 3.67      | 0.8              | 2.296607      | 0.822222      | 0.818731 |
| 0.00500 | 3.67      | 1.2              | 0.855941      | 0.691852      | 0.707454 |
| 0.00500 | 0.50      | 0.8              | 2.296607      | 0.754815      | 0.700452 |
| 0.00500 | 3.67      | 1.5              | 0.507459      | 0.711111      | 0.699074 |
| 0.00010 | 0.50      | 1.2              | 0.855941      | 0.518519      | 0.585459 |
| 0.00010 | 0.50      | 0.8              | 2.296607      | 0.491852      | 0.584242 |
| 0.00010 | 3.67      | 1.5              | 0.507459      | 0.528889      | 0.577689 |
| 0.00005 | 1.00      | 1.5              | 0.507459      | 0.522963      | 0.548387 |
| 0.00005 | 3.67      | 0.8              | 2.296607      | 0.534815      | 0.517665 |
| 0.00500 | 1.00      | 0.8              | 2.296607      | 0.639259      | 0.508577 |
| 0.00500 | 1.00      | 1.5              | 0.507459      | 0.554074      | 0.453721 |
| 0.00005 | 3.67      | 1.2              | 0.855941      | 0.525185      | 0.447890 |
| 0.00005 | 0.50      | 0.8              | 2.296607      | 0.506667      | 0.445923 |
| 0.00010 | 3.67      | 0.8              | 2.296607      | 0.552593      | 0.424762 |

```

0.00010      3.67      1.2      0.855941      0.500741  0.419966
def train_baseline_robust(config): # for storing epoch wise scores (takes longer)
    print("\n" + "="*50 + "\nRunning Improved Non-Private Baseline Training...\n" + "="*50)
    train_loader = DataLoader(train_dataset, batch_size=config['batch_size'], shuffle=True, collate_fn=collate_bow)
    val_loader = DataLoader(val_dataset, batch_size=config['batch_size'], collate_fn=collate_bow)
    test_loader = DataLoader(test_dataset, batch_size=config['batch_size'], collate_fn=collate_bow)
    model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(gpu_device)
    optimizer = AdamW(model.parameters(), lr=config['lr'])
    scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='max', factor=0.1, patience=3)
    criterion = nn.BCEWithLogitsLoss()
    best_val_f1 = -1
    patience_counter = 0
    best_model_state = None
    history = []
    for epoch in range(config['epochs']):
        model.train()
        for texts, offsets, cats, labels in tqdm(train_loader, desc=f"Epoch {epoch+1}/{config['epochs']}"):
            texts, offsets, cats, labels = texts.to(gpu_device), offsets.to(gpu_device), cats.to(gpu_device), labels.to(gpu_device)
            optimizer.zero_grad()
            logits = model(texts, offsets, cats)
            loss = criterion(logits, labels)
            loss.backward()
            optimizer.step()

        val_metrics = eval_model(model, val_loader, gpu_device)
        val_f1 = val_metrics['f1']
        history.append({'epoch': epoch + 1, 'val_accuracy': val_metrics['accuracy'], 'val_f1': val_f1})

        print(f"Epoch {epoch+1}: Val F1: {val_f1:.4f}, Current LR: {optimizer.param_groups[0]['lr']:.2E}")
        scheduler.step(val_f1)
        if val_f1 > best_val_f1:
            best_val_f1 = val_f1
            patience_counter = 0
            best_model_state = copy.deepcopy(model.state_dict())
        else:
            patience_counter += 1
            if patience_counter >= config['patience']:
                print(f"--- Early stopping triggered. ---")
                break
    print("\nLoading best model and evaluating on the test set...")
    model.load_state_dict(best_model_state)
    test_metrics = eval_model(model, test_loader, gpu_device)
    print(f"\nIMPROVED BASELINE RESULTS:\n -> Final Test Accuracy: {test_metrics['accuracy']:.4f}\n -> Final Test F1: {val_f1:.4f}")
    return test_metrics, pd.DataFrame(history)

if __name__ == '__main__':
    baseline_config = {'lr': 1e-3, 'batch_size': 128, 'epochs': 50, 'patience': 5}
    baseline_metrics, baseline_history = train_baseline_robust(baseline_config)

    DP_BATCH_SIZE = 256
    DP_EPOCHS = 15

    temp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
    temp_loader = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
    adaptive_c = find_adaptive_clip_norm(temp_model, temp_loader, cpu_device)

    CLIP_NORMS = sorted(list(set([0.5, 1.0, round(adaptive_c, 2)])))
    NOISE_MULTIPLIERS = [0.2, 0.4, 0.8]
    LEARNING_RATES = [1e-3, 5e-3]

    print(f"\nStarting grid sweep with C values: {CLIP_NORMS}")

    results = []
    all_dp_histories = {}

    for lr in LEARNING_RATES:
        for c in CLIP_NORMS:
            for sigma in NOISE_MULTIPLIERS:
                run_id = f"C={c}, σ={sigma}, LR={lr}"
                print("\n" + "-" * 50 + f"\nStarting DP-SGD Sweep Run: {run_id}")

                dp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
                dp_optimizer = AdamW(dp_model.parameters(), lr=lr)
                dp_criterion = nn.BCEWithLogitsLoss()

                train_loader_cpu = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)

```

```

test_loader_cpu = DataLoader(test_dataset, batch_size=DP_BATCH_SIZE, collate_fn=collate_bow)

accountant = RDPAccountant()
DELTA = 1 / len(train_dataset)
sample_rate = DP_BATCH_SIZE / len(train_dataset)

history = []
for epoch in range(DP_EPOCHS):
    train_manual_dp(dp_model, train_loader_cpu, dp_optimizer, dp_criterion, cpu_device, c, sigma)
    accountant.step(noise_multiplier=sigma, sample_rate=sample_rate)
    epsilon = accountant.get_epsilon(delta=DELTA)

    epoch_metrics = eval_model(dp_model, test_loader_cpu, cpu_device)
    history.append({
        'epoch': epoch + 1,
        'accuracy': epoch_metrics['accuracy'],
        'epsilon': epsilon
    })
    print(f"Epoch {epoch+1}/{DP_EPOCHS} | Epsilon: {epsilon:.2f} | Test Accuracy: {epoch_metrics['accuracy']:.4f}")

all_dp_histories[run_id] = pd.DataFrame(history)
final_metrics = history[-1]

results.append({
    "run_id": run_id, "lr": lr, "clip_norm": c, "noise_multiplier": sigma,
    "final_epsilon": final_metrics['epsilon'], "accuracy": final_metrics['accuracy'], "f1": eval_model(
    })

results_df = pd.DataFrame(results).rename(columns={"accuracy": "test_accuracy", "f1": "test_f1"})
display_cols = ['run_id', 'lr', 'clip_norm', 'noise_multiplier', 'final_epsilon', 'test_accuracy', 'test_f1']
results_df = results_df[display_cols].sort_values(by="test_f1", ascending=False)

print("\n\n" + "=" * 80 + "\n          Final Tuned Sweep Results: Baseline vs. DP-SGD\n" + "=" * 80)
print(f"BASELINE MODEL (Bag-of-Words, Non-Private):")
print(f"  Accuracy: {baseline_metrics['accuracy']:.4f}, F1 Score: {baseline_metrics['f1']:.4f}\n")
print(f"DP-SGD MODELS (Bag-of-Words, Private,  $\delta \approx$  {DELTA:.2E}):")
print(results_df.to_string(index=False))
print("=" * 80)

print("\n\n" + "=" * 80 + "\n          Plotting Top 5 DP-SGD Runs...\n" + "=" * 80)

top_5_runs = results_df.head(5)

fig, ax1 = plt.subplots(figsize=(14, 8))
colors = plt.cm.viridis(np.linspace(0, 1, len(top_5_runs)))

ax1.plot(baseline_history['epoch'], baseline_history['val_accuracy'],
        label='Baseline (Val Accuracy)', color='black', linestyle='--', linewidth=2.5)

for i, (idx, row) in enumerate(top_5_runs.iterrows()):
    run_id = row['run_id']
    history_df = all_dp_histories[run_id]
    ax1.plot(history_df['epoch'], history_df['accuracy'], label=f'{run_id} (Accuracy)', color=colors[i], linewidth=2)

ax1.set_xlabel('Epoch')
ax1.set_ylabel('Test Accuracy')
ax1.set_title('DP-SGD Model Performance: Accuracy and Privacy Loss ( $\epsilon$ ) vs. Epoch')
ax1.grid(True)
ax1.set_ylim(bottom=0.45)

ax2 = ax1.twinx()
for i, (idx, row) in enumerate(top_5_runs.iterrows()):
    run_id = row['run_id']
    history_df = all_dp_histories[run_id]
    ax2.plot(history_df['epoch'], history_df['epsilon'], linestyle=':', marker='o', markersize=4, color=colors[i],
            label=f'{run_id} (Epsilon)')

ax2.set_ylabel('Epsilon ( $\epsilon$ )')

lines, labels = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax2.legend(lines + lines2, labels + labels2, loc='best')

fig.tight_layout()
plt.savefig("dp_sgd_learning_curves.png")
print("\nPlot saved to dp_sgd_learning_curves.png")
plt.show()

```



```
Using device for Baseline: cuda
Using device for DP-SGD: cpu
```

```
=====
Running Improved Non-Private Baseline Training...
=====
```

```
Epoch 1/50: 100%                               50/50 [00:00<00:00, 236.42it/s]
Epoch 1: Val F1: 0.7527, Current LR: 1.00E-03
Epoch 2/50: 100%                               50/50 [00:00<00:00, 252.61it/s]
Epoch 2: Val F1: 0.9157, Current LR: 1.00E-03
Epoch 3/50: 100%                               50/50 [00:00<00:00, 248.58it/s]
Epoch 3: Val F1: 0.9870, Current LR: 1.00E-03
Epoch 4/50: 100%                               50/50 [00:00<00:00, 259.57it/s]
Epoch 4: Val F1: 0.9964, Current LR: 1.00E-03
Epoch 5/50: 100%                               50/50 [00:00<00:00, 252.19it/s]
Epoch 5: Val F1: 0.9986, Current LR: 1.00E-03
Epoch 6/50: 100%                               50/50 [00:00<00:00, 259.87it/s]
Epoch 6: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 7/50: 100%                               50/50 [00:00<00:00, 243.03it/s]
Epoch 7: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 8/50: 100%                               50/50 [00:00<00:00, 255.99it/s]
Epoch 8: Val F1: 1.0000, Current LR: 1.00E-03
Epoch 9/50: 100%                               50/50 [00:00<00:00, 256.05it/s]
Epoch 9: Val F1: 1.0000, Current LR: 1.00E-03
Epoch 10/50: 100%                              50/50 [00:00<00:00, 247.73it/s]
Epoch 10: Val F1: 1.0000, Current LR: 1.00E-03
Epoch 11/50: 100%                              50/50 [00:00<00:00, 242.61it/s]
Epoch 11: Val F1: 1.0000, Current LR: 1.00E-03
Epoch 12/50: 100%                              50/50 [00:00<00:00, 214.63it/s]
Epoch 12: Val F1: 1.0000, Current LR: 1.00E-03
Epoch 13/50: 100%                              50/50 [00:00<00:00, 238.63it/s]
Epoch 13: Val F1: 1.0000, Current LR: 1.00E-04
--- Early stopping triggered. ---
```

```
Loading best model and evaluating on the test set...
```

```
IMPROVED BASELINE RESULTS:
```

```
-> Final Test Accuracy: 1.0000
-> Final Test F1 Score: 1.0000
```

```
--- Running epoch to find adaptive clipping norm ---
```

```
Analyzing Gradients: 100%                        25/25 [00:07<00:00, 3.98it/s]
```

```
--- Found median gradient norm: 3.3147 ---
```

```
Starting grid sweep with C values: [0.5, 1.0, np.float64(3.31)]
```

```
-----
Starting DP-SGD Sweep Run: C=0.5,  $\sigma=0.2$ , LR=0.001
```

```
Manual DP Batch: 100%                          25/25 [00:14<00:00, 2.11it/s]
Epoch 1/15 | Epsilon: 24.28 | Test Accuracy: 0.5326
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.10it/s]
Epoch 2/15 | Epsilon: 32.72 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 1.89it/s]
Epoch 3/15 | Epsilon: 37.76 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.09it/s]
Epoch 4/15 | Epsilon: 41.40 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:14<00:00, 1.77it/s]
Epoch 5/15 | Epsilon: 45.05 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 1.88it/s]
Epoch 6/15 | Epsilon: 48.69 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.02it/s]
Epoch 7/15 | Epsilon: 52.34 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.10it/s]
Epoch 8/15 | Epsilon: 54.00 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.14it/s]
```



Epoch 9/15 | Epsilon: 55.62 | Test Accuracy: 0.5111  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.14it/s]  
Epoch 10/15 | Epsilon: 57.24 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 11/15 | Epsilon: 58.86 | Test Accuracy: 0.5067  
Manual DP Batch: 100% 25/25 [00:15<00:00, 1.57it/s]  
Epoch 12/15 | Epsilon: 60.48 | Test Accuracy: 0.5170  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.82it/s]  
Epoch 13/15 | Epsilon: 62.10 | Test Accuracy: 0.5489  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.90it/s]  
Epoch 14/15 | Epsilon: 63.72 | Test Accuracy: 0.6363  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.02it/s]  
Epoch 15/15 | Epsilon: 65.34 | Test Accuracy: 0.7563

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =0.4, LR=0.001  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 1/15 | Epsilon: 7.14 | Test Accuracy: 0.5526  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.12it/s]  
Epoch 2/15 | Epsilon: 8.40 | Test Accuracy: 0.5096  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.14it/s]  
Epoch 3/15 | Epsilon: 9.21 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.16it/s]  
Epoch 4/15 | Epsilon: 9.83 | Test Accuracy: 0.5311  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.12it/s]  
Epoch 5/15 | Epsilon: 10.38 | Test Accuracy: 0.5348  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 6/15 | Epsilon: 10.81 | Test Accuracy: 0.5222  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.84it/s]  
Epoch 7/15 | Epsilon: 11.23 | Test Accuracy: 0.5007  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.85it/s]  
Epoch 8/15 | Epsilon: 11.66 | Test Accuracy: 0.5207  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.65it/s]  
Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.5141  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.04it/s]  
Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.5178  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.5015  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.5022  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.5015  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.5030  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.14it/s]  
Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.5030

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =0.8, LR=0.001  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.98it/s]  
Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.5296  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.76it/s]  
Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.5052  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.88it/s]  
Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.5393  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.02it/s]  
Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.5178  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.09it/s]  
Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.5289  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.5533  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.12it/s]

|                                                           |                               |
|-----------------------------------------------------------|-------------------------------|
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 7/15   Epsilon: 2.24   Test Accuracy: 0.5615        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 8/15   Epsilon: 2.30   Test Accuracy: 0.5104        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.90it/s] |
| Epoch 9/15   Epsilon: 2.35   Test Accuracy: 0.5178        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.72it/s] |
| Epoch 10/15   Epsilon: 2.39   Test Accuracy: 0.5059       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.90it/s] |
| Epoch 11/15   Epsilon: 2.44   Test Accuracy: 0.5185       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.06it/s] |
| Epoch 12/15   Epsilon: 2.48   Test Accuracy: 0.5030       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 13/15   Epsilon: 2.52   Test Accuracy: 0.5163       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.71it/s] |
| Epoch 14/15   Epsilon: 2.56   Test Accuracy: 0.5133       |                               |
| Manual DP Batch: 100%                                     |                               |
| Epoch 15/15   Epsilon: 2.59   Test Accuracy: 0.5126       |                               |
| -----                                                     |                               |
| Starting DP-SGD Sweep Run: C=1.0, $\sigma$ =0.2, LR=0.001 |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 1/15   Epsilon: 24.28   Test Accuracy: 0.5000       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.16it/s] |
| Epoch 2/15   Epsilon: 32.72   Test Accuracy: 0.5000       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 3/15   Epsilon: 37.76   Test Accuracy: 0.5015       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.91it/s] |
| Epoch 4/15   Epsilon: 41.40   Test Accuracy: 0.5748       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.69it/s] |
| Epoch 5/15   Epsilon: 45.05   Test Accuracy: 0.5030       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.93it/s] |
| Epoch 6/15   Epsilon: 48.69   Test Accuracy: 0.5000       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.03it/s] |
| Epoch 7/15   Epsilon: 52.34   Test Accuracy: 0.5000       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.07it/s] |
| Epoch 8/15   Epsilon: 54.00   Test Accuracy: 0.5259       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 9/15   Epsilon: 55.62   Test Accuracy: 0.5052       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 10/15   Epsilon: 57.24   Test Accuracy: 0.5148      |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 2.12it/s] |
| Epoch 11/15   Epsilon: 58.86   Test Accuracy: 0.5341      |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 12/15   Epsilon: 60.48   Test Accuracy: 0.5563      |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.86it/s] |
| Epoch 13/15   Epsilon: 62.10   Test Accuracy: 0.5726      |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.74it/s] |
| Epoch 14/15   Epsilon: 63.72   Test Accuracy: 0.6719      |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.97it/s] |
| Epoch 15/15   Epsilon: 65.34   Test Accuracy: 0.7304      |                               |
| -----                                                     |                               |
| Starting DP-SGD Sweep Run: C=1.0, $\sigma$ =0.4, LR=0.001 |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.04it/s] |
| Epoch 1/15   Epsilon: 7.14   Test Accuracy: 0.5111        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 2/15   Epsilon: 8.40   Test Accuracy: 0.5133        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 3/15   Epsilon: 9.21   Test Accuracy: 0.5889        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.13it/s] |
| Epoch 4/15   Epsilon: 9.83   Test Accuracy: 0.5474        |                               |

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 5/15 | Epsilon: 10.38 | Test Accuracy: 0.5415

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 6/15 | Epsilon: 10.81 | Test Accuracy: 0.5526

Manual DP Batch: 100%25/25 [00:14<00:00, 2.11it/s]

Epoch 7/15 | Epsilon: 11.23 | Test Accuracy: 0.5526

Manual DP Batch: 100%25/25 [00:14<00:00, 1.90it/s]

Epoch 8/15 | Epsilon: 11.66 | Test Accuracy: 0.5533

Manual DP Batch: 100%25/25 [00:14<00:00, 1.74it/s]

Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.5341

Manual DP Batch: 100%25/25 [00:13<00:00, 1.90it/s]

Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.5193

Manual DP Batch: 100%25/25 [00:13<00:00, 2.03it/s]

Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.5200

Manual DP Batch: 100%25/25 [00:13<00:00, 2.07it/s]

Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.5170

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.5089

Manual DP Batch: 100%25/25 [00:14<00:00, 2.10it/s]

Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.5281

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.5793

-----

Starting DP-SGD Sweep Run: C=1.0,  $\sigma$ =0.8, LR=0.001

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:14<00:00, 1.74it/s]

Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 1.89it/s]

Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 1.99it/s]

Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.05it/s]

Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 7/15 | Epsilon: 2.24 | Test Accuracy: 0.5393

Manual DP Batch: 100%25/25 [00:13<00:00, 2.15it/s]

Epoch 8/15 | Epsilon: 2.30 | Test Accuracy: 0.5252

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 9/15 | Epsilon: 2.35 | Test Accuracy: 0.5622

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 10/15 | Epsilon: 2.39 | Test Accuracy: 0.5896

Manual DP Batch: 100%25/25 [00:13<00:00, 1.89it/s]

Epoch 11/15 | Epsilon: 2.44 | Test Accuracy: 0.5274

Manual DP Batch: 100%25/25 [00:14<00:00, 1.71it/s]

Epoch 12/15 | Epsilon: 2.48 | Test Accuracy: 0.5533

Manual DP Batch: 100%25/25 [00:14<00:00, 1.98it/s]

Epoch 13/15 | Epsilon: 2.52 | Test Accuracy: 0.5704

Manual DP Batch: 100%25/25 [00:13<00:00, 2.07it/s]

Epoch 14/15 | Epsilon: 2.56 | Test Accuracy: 0.5452

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 15/15 | Epsilon: 2.59 | Test Accuracy: 0.5126

-----

Starting DP-SGD Sweep Run: C=3.31,  $\sigma$ =0.2, LR=0.001

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 1/15 | Epsilon: 24.28 | Test Accuracy: 0.5719

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 2/15 | Epsilon: 22.72 | Test Accuracy: 0.5341

```
Epoch 2/15 | Epsilon: 32.72 | Test Accuracy: 0.5541
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.14it/s]
Epoch 3/15 | Epsilon: 37.76 | Test Accuracy: 0.6304
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]
Epoch 4/15 | Epsilon: 41.40 | Test Accuracy: 0.5459
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.73it/s]
Epoch 5/15 | Epsilon: 45.05 | Test Accuracy: 0.5600
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.84it/s]
Epoch 6/15 | Epsilon: 48.69 | Test Accuracy: 0.6430
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.00it/s]
Epoch 7/15 | Epsilon: 52.34 | Test Accuracy: 0.6407
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]
Epoch 8/15 | Epsilon: 54.00 | Test Accuracy: 0.6822
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.10it/s]
Epoch 9/15 | Epsilon: 55.62 | Test Accuracy: 0.7504
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.08it/s]
Epoch 10/15 | Epsilon: 57.24 | Test Accuracy: 0.7622
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.09it/s]
Epoch 11/15 | Epsilon: 58.86 | Test Accuracy: 0.7881
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.02it/s]
Epoch 12/15 | Epsilon: 60.48 | Test Accuracy: 0.8022
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.71it/s]
Epoch 13/15 | Epsilon: 62.10 | Test Accuracy: 0.7800
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.84it/s]
Epoch 14/15 | Epsilon: 63.72 | Test Accuracy: 0.8096
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.93it/s]
Epoch 15/15 | Epsilon: 65.34 | Test Accuracy: 0.7904

-----
Starting DP-SGD Sweep Run: C=3.31,  $\sigma=0.4$ , LR=0.001
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]
Epoch 1/15 | Epsilon: 7.14 | Test Accuracy: 0.5363
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]
Epoch 2/15 | Epsilon: 8.40 | Test Accuracy: 0.5748
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]
Epoch 3/15 | Epsilon: 9.21 | Test Accuracy: 0.6274
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.98it/s]
Epoch 4/15 | Epsilon: 9.83 | Test Accuracy: 0.5985
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.93it/s]
Epoch 5/15 | Epsilon: 10.38 | Test Accuracy: 0.6200
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.71it/s]
Epoch 6/15 | Epsilon: 10.81 | Test Accuracy: 0.5793
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.95it/s]
Epoch 7/15 | Epsilon: 11.23 | Test Accuracy: 0.5867
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.01it/s]
Epoch 8/15 | Epsilon: 11.66 | Test Accuracy: 0.5837
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]
Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.6007
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]
Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.6052
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]
Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.6015
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]
Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.6067
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.92it/s]
Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.6326
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.69it/s]
Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.7163
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.98it/s]
Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.6733

-----
```

Starting DP-SGD Sweep Run: C=3.31,  $\sigma=0.8$ , LR=0.001

Manual DP Batch: 100%25/25 [00:14<00:00, 2.05it/s]

Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.5444

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.5659

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.5667

Manual DP Batch: 100%25/25 [00:13<00:00, 2.08it/s]

Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.5763

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.5667

Manual DP Batch: 100%25/25 [00:14<00:00, 1.85it/s]

Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.5844

Manual DP Batch: 100%25/25 [00:14<00:00, 1.78it/s]

Epoch 7/15 | Epsilon: 2.24 | Test Accuracy: 0.6074

Manual DP Batch: 100%25/25 [00:13<00:00, 1.94it/s]

Epoch 8/15 | Epsilon: 2.30 | Test Accuracy: 0.5430

Manual DP Batch: 100%25/25 [00:13<00:00, 2.05it/s]

Epoch 9/15 | Epsilon: 2.35 | Test Accuracy: 0.5496

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 10/15 | Epsilon: 2.39 | Test Accuracy: 0.5504

Manual DP Batch: 100%25/25 [00:14<00:00, 2.11it/s]

Epoch 11/15 | Epsilon: 2.44 | Test Accuracy: 0.5370

Manual DP Batch: 100%25/25 [00:13<00:00, 2.05it/s]

Epoch 12/15 | Epsilon: 2.48 | Test Accuracy: 0.5156

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 13/15 | Epsilon: 2.52 | Test Accuracy: 0.5096

Manual DP Batch: 100%25/25 [00:14<00:00, 1.89it/s]

Epoch 14/15 | Epsilon: 2.56 | Test Accuracy: 0.5104

Manual DP Batch: 100%25/25 [00:14<00:00, 1.69it/s]

Epoch 15/15 | Epsilon: 2.59 | Test Accuracy: 0.5081

-----

Starting DP-SGD Sweep Run: C=0.5,  $\sigma=0.2$ , LR=0.005

Manual DP Batch: 100%25/25 [00:13<00:00, 1.94it/s]

Epoch 1/15 | Epsilon: 24.28 | Test Accuracy: 0.5052

Manual DP Batch: 100%25/25 [00:13<00:00, 2.04it/s]

Epoch 2/15 | Epsilon: 32.72 | Test Accuracy: 0.5000

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 3/15 | Epsilon: 37.76 | Test Accuracy: 0.5370

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 4/15 | Epsilon: 41.40 | Test Accuracy: 0.6578

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 5/15 | Epsilon: 45.05 | Test Accuracy: 0.7015

Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 6/15 | Epsilon: 48.69 | Test Accuracy: 0.8378

Manual DP Batch: 100%25/25 [00:14<00:00, 2.13it/s]

Epoch 7/15 | Epsilon: 52.34 | Test Accuracy: 0.7778

Manual DP Batch: 100%25/25 [00:14<00:00, 2.12it/s]

Epoch 8/15 | Epsilon: 54.00 | Test Accuracy: 0.8844

Manual DP Batch: 100%25/25 [00:13<00:00, 1.88it/s]

Epoch 9/15 | Epsilon: 55.62 | Test Accuracy: 0.9037

Manual DP Batch: 100%25/25 [00:14<00:00, 1.73it/s]

Epoch 10/15 | Epsilon: 57.24 | Test Accuracy: 0.9111

Manual DP Batch: 100%25/25 [00:13<00:00, 1.96it/s]

Epoch 11/15 | Epsilon: 58.86 | Test Accuracy: 0.9393

Manual DP Batch: 100%25/25 [00:13<00:00, 2.03it/s]

Epoch 12/15 | Epsilon: 60.48 | Test Accuracy: 0.9393

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 13/15 | Epsilon: 62.10 | Test Accuracy: 0.9481

Manual DP Batch: 100%25/25 [00:13<00:00, 2.16it/s]

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.13it/s]

Epoch 14/15 | Epsilon: 63.72 | Test Accuracy: 0.9570

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.11it/s]

Epoch 15/15 | Epsilon: 65.34 | Test Accuracy: 0.9556

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =0.4, LR=0.005

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.10it/s]

Epoch 1/15 | Epsilon: 7.14 | Test Accuracy: 0.5430

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.12it/s]

Epoch 2/15 | Epsilon: 8.40 | Test Accuracy: 0.5844

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 2.11it/s]

Epoch 3/15 | Epsilon: 9.21 | Test Accuracy: 0.5000

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 1.89it/s]

Epoch 4/15 | Epsilon: 9.83 | Test Accuracy: 0.5281

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 1.76it/s]

Epoch 5/15 | Epsilon: 10.38 | Test Accuracy: 0.5807

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 1.94it/s]

Epoch 6/15 | Epsilon: 10.81 | Test Accuracy: 0.7615

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 2.06it/s]

Epoch 7/15 | Epsilon: 11.23 | Test Accuracy: 0.8015

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.06it/s]

Epoch 8/15 | Epsilon: 11.66 | Test Accuracy: 0.8148

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.14it/s]

Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.8244

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.12it/s]

Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.8289

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.11it/s]

Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.8637

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.11it/s]

Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.8807

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 1.85it/s]

Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.8963

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 1.90it/s]

Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.9007

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.08it/s]

Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.8948

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =0.8, LR=0.005

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.07it/s]

Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.4985

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.10it/s]

Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.6037

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.12it/s]

Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.5074

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.14it/s]

Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.5111

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 2.11it/s]

Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.5081

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 1.93it/s]

Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.5044

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 1.75it/s]

Epoch 7/15 | Epsilon: 2.24 | Test Accuracy: 0.5585

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 1.86it/s]

Epoch 8/15 | Epsilon: 2.30 | Test Accuracy: 0.6370

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.02it/s]

Epoch 9/15 | Epsilon: 2.35 | Test Accuracy: 0.7711

Manual DP Batch: 100%

25/25 [00:14&lt;00:00, 2.07it/s]

Epoch 10/15 | Epsilon: 2.39 | Test Accuracy: 0.7044

Manual DP Batch: 100%

25/25 [00:13&lt;00:00, 2.12it/s]

Epoch 11/15 | Epsilon: 2.44 | Test Accuracy: 0.7444

|                                                     |                               |
|-----------------------------------------------------|-------------------------------|
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 12/15   Epsilon: 2.48   Test Accuracy: 0.7681 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 13/15   Epsilon: 2.52   Test Accuracy: 0.8096 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 14/15   Epsilon: 2.56   Test Accuracy: 0.8222 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.08it/s] |
| Epoch 15/15   Epsilon: 2.59   Test Accuracy: 0.7778 |                               |

-----  
Starting DP-SGD Sweep Run: C=1.0,  $\sigma=0.2$ , LR=0.005

|                                                      |                               |
|------------------------------------------------------|-------------------------------|
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 1.86it/s] |
| Epoch 1/15   Epsilon: 24.28   Test Accuracy: 0.4985  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 1.79it/s] |
| Epoch 2/15   Epsilon: 32.72   Test Accuracy: 0.5000  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 1.95it/s] |
| Epoch 3/15   Epsilon: 37.76   Test Accuracy: 0.6022  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.04it/s] |
| Epoch 4/15   Epsilon: 41.40   Test Accuracy: 0.7148  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 5/15   Epsilon: 45.05   Test Accuracy: 0.7400  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:14<00:00, 1.99it/s] |
| Epoch 6/15   Epsilon: 48.69   Test Accuracy: 0.7578  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 7/15   Epsilon: 52.34   Test Accuracy: 0.8370  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 8/15   Epsilon: 54.00   Test Accuracy: 0.8481  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.10it/s] |
| Epoch 9/15   Epsilon: 55.62   Test Accuracy: 0.8919  |                               |
| Manual DP Batch: 100%                                | 25/25 [00:14<00:00, 1.74it/s] |
| Epoch 10/15   Epsilon: 57.24   Test Accuracy: 0.9007 |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 1.87it/s] |
| Epoch 11/15   Epsilon: 58.86   Test Accuracy: 0.9193 |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.00it/s] |
| Epoch 12/15   Epsilon: 60.48   Test Accuracy: 0.9096 |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 13/15   Epsilon: 62.10   Test Accuracy: 0.9393 |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 14/15   Epsilon: 63.72   Test Accuracy: 0.9422 |                               |
| Manual DP Batch: 100%                                | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 15/15   Epsilon: 65.34   Test Accuracy: 0.9444 |                               |

-----  
Starting DP-SGD Sweep Run: C=1.0,  $\sigma=0.4$ , LR=0.005

|                                                     |                               |
|-----------------------------------------------------|-------------------------------|
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 1/15   Epsilon: 7.14   Test Accuracy: 0.5000  |                               |
| Manual DP Batch: 100%                               | 25/25 [00:14<00:00, 1.82it/s] |
| Epoch 2/15   Epsilon: 8.40   Test Accuracy: 0.5000  |                               |
| Manual DP Batch: 100%                               | 25/25 [00:14<00:00, 1.91it/s] |
| Epoch 3/15   Epsilon: 9.21   Test Accuracy: 0.5007  |                               |
| Manual DP Batch: 100%                               | 25/25 [00:14<00:00, 1.67it/s] |
| Epoch 4/15   Epsilon: 9.83   Test Accuracy: 0.5948  |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 1.92it/s] |
| Epoch 5/15   Epsilon: 10.38   Test Accuracy: 0.6600 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.06it/s] |
| Epoch 6/15   Epsilon: 10.81   Test Accuracy: 0.7481 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 7/15   Epsilon: 11.23   Test Accuracy: 0.7807 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 8/15   Epsilon: 11.66   Test Accuracy: 0.8000 |                               |
| Manual DP Batch: 100%                               | 25/25 [00:13<00:00, 2.16it/s] |

Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.8193  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.16it/s]  
Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.8259  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.16it/s]  
Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.8578  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.16it/s]  
Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.8541  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.03it/s]  
Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.8637  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.70it/s]  
Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.8919  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.93it/s]  
Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.9059

-----  
Starting DP-SGD Sweep Run: C=1.0,  $\sigma$ =0.8, LR=0.005  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.04it/s]  
Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.08it/s]  
Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.11it/s]  
Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.13it/s]  
Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.5296  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.5044  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.87it/s]  
Epoch 7/15 | Epsilon: 2.24 | Test Accuracy: 0.5985  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.71it/s]  
Epoch 8/15 | Epsilon: 2.30 | Test Accuracy: 0.6444  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.92it/s]  
Epoch 9/15 | Epsilon: 2.35 | Test Accuracy: 0.7111  
Manual DP Batch: 100% 25/25 [00:19<00:00, 1.69it/s]  
Epoch 10/15 | Epsilon: 2.39 | Test Accuracy: 0.7452  
Manual DP Batch: 100% 25/25 [00:16<00:00, 1.61it/s]  
Epoch 11/15 | Epsilon: 2.44 | Test Accuracy: 0.7496  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.84it/s]  
Epoch 12/15 | Epsilon: 2.48 | Test Accuracy: 0.7859  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.81it/s]  
Epoch 13/15 | Epsilon: 2.52 | Test Accuracy: 0.7926  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.95it/s]  
Epoch 14/15 | Epsilon: 2.56 | Test Accuracy: 0.7541  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.03it/s]  
Epoch 15/15 | Epsilon: 2.59 | Test Accuracy: 0.7615

-----  
Starting DP-SGD Sweep Run: C=3.31,  $\sigma$ =0.2, LR=0.005  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.07it/s]  
Epoch 1/15 | Epsilon: 24.28 | Test Accuracy: 0.5215  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]  
Epoch 2/15 | Epsilon: 32.72 | Test Accuracy: 0.5800  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 3/15 | Epsilon: 37.76 | Test Accuracy: 0.7096  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 4/15 | Epsilon: 41.40 | Test Accuracy: 0.8437  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]  
Epoch 5/15 | Epsilon: 45.05 | Test Accuracy: 0.8459  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.72it/s]  
Epoch 6/15 | Epsilon: 48.69 | Test Accuracy: 0.8622  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.90it/s]



Epoch 7/15 | Epsilon: 52.34 | Test Accuracy: 0.8985  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.05it/s]  
Epoch 8/15 | Epsilon: 54.00 | Test Accuracy: 0.9244  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.07it/s]  
Epoch 9/15 | Epsilon: 55.62 | Test Accuracy: 0.9341  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 10/15 | Epsilon: 57.24 | Test Accuracy: 0.9407  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 11/15 | Epsilon: 58.86 | Test Accuracy: 0.9519  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.12it/s]  
Epoch 12/15 | Epsilon: 60.48 | Test Accuracy: 0.9526  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.09it/s]  
Epoch 13/15 | Epsilon: 62.10 | Test Accuracy: 0.9644  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.89it/s]  
Epoch 14/15 | Epsilon: 63.72 | Test Accuracy: 0.9615  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.79it/s]  
Epoch 15/15 | Epsilon: 65.34 | Test Accuracy: 0.9674

-----  
Starting DP-SGD Sweep Run: C=3.31,  $\sigma$ =0.4, LR=0.005  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.94it/s]  
Epoch 1/15 | Epsilon: 7.14 | Test Accuracy: 0.6185  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.67it/s]  
Epoch 2/15 | Epsilon: 8.40 | Test Accuracy: 0.6081  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.08it/s]  
Epoch 3/15 | Epsilon: 9.21 | Test Accuracy: 0.6667  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.04it/s]  
Epoch 4/15 | Epsilon: 9.83 | Test Accuracy: 0.7244  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.13it/s]  
Epoch 5/15 | Epsilon: 10.38 | Test Accuracy: 0.7274  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.04it/s]  
Epoch 6/15 | Epsilon: 10.81 | Test Accuracy: 0.8170  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.79it/s]  
Epoch 7/15 | Epsilon: 11.23 | Test Accuracy: 0.8326  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.85it/s]  
Epoch 8/15 | Epsilon: 11.66 | Test Accuracy: 0.8385  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.97it/s]  
Epoch 9/15 | Epsilon: 12.04 | Test Accuracy: 0.8252  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]  
Epoch 10/15 | Epsilon: 12.33 | Test Accuracy: 0.8637  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.09it/s]  
Epoch 11/15 | Epsilon: 12.63 | Test Accuracy: 0.8667  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.10it/s]  
Epoch 12/15 | Epsilon: 12.92 | Test Accuracy: 0.8830  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.11it/s]  
Epoch 13/15 | Epsilon: 13.22 | Test Accuracy: 0.8711  
Manual DP Batch: 100% 25/25 [00:14<00:00, 2.10it/s]  
Epoch 14/15 | Epsilon: 13.51 | Test Accuracy: 0.8904  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.86it/s]  
Epoch 15/15 | Epsilon: 13.80 | Test Accuracy: 0.9015

-----  
Starting DP-SGD Sweep Run: C=3.31,  $\sigma$ =0.8, LR=0.005  
Manual DP Batch: 100% 25/25 [00:14<00:00, 1.76it/s]  
Epoch 1/15 | Epsilon: 1.73 | Test Accuracy: 0.5578  
Manual DP Batch: 100% 25/25 [00:13<00:00, 1.93it/s]  
Epoch 2/15 | Epsilon: 1.87 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.07it/s]  
Epoch 3/15 | Epsilon: 1.98 | Test Accuracy: 0.5000  
Manual DP Batch: 100% 25/25 [00:13<00:00, 2.06it/s]  
Epoch 4/15 | Epsilon: 2.06 | Test Accuracy: 0.5704

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 5/15 | Epsilon: 2.13 | Test Accuracy: 0.6104

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 6/15 | Epsilon: 2.19 | Test Accuracy: 0.6259

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 7/15 | Epsilon: 2.24 | Test Accuracy: 0.6889

Manual DP Batch: 100%25/25 [00:13<00:00, 2.00it/s]

Epoch 8/15 | Epsilon: 2.30 | Test Accuracy: 0.7030

Manual DP Batch: 100%25/25 [00:14<00:00, 1.73it/s]

Epoch 9/15 | Epsilon: 2.35 | Test Accuracy: 0.6852

Manual DP Batch: 100%25/25 [00:14<00:00, 1.66it/s]

Epoch 10/15 | Epsilon: 2.39 | Test Accuracy: 0.7600

Manual DP Batch: 100%25/25 [00:14<00:00, 2.05it/s]

Epoch 11/15 | Epsilon: 2.44 | Test Accuracy: 0.7415

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 12/15 | Epsilon: 2.48 | Test Accuracy: 0.7511

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 13/15 | Epsilon: 2.52 | Test Accuracy: 0.8074

Manual DP Batch: 100%25/25 [00:14<00:00, 2.09it/s]

Epoch 14/15 | Epsilon: 2.56 | Test Accuracy: 0.8037

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 15/15 | Epsilon: 2.59 | Test Accuracy: 0.7778

Final Tuned Sweep Results: Baseline vs. DP-SGD

BASELINE MODEL (Bag-of-Words, Non-Private):

Accuracy: 1.0000, F1 Score: 1.0000

DP-SGD MODELS (Bag-of-Words, Private,  $\delta \approx 1.59E-04$ ):

|                                 | run_id | lr   | clip_norm | noise_multiplier | final_epsilon | test_accuracy | test_f1 |
|---------------------------------|--------|------|-----------|------------------|---------------|---------------|---------|
| C=3.31, $\sigma=0.2$ , LR=0.005 | 0.005  | 3.31 | 0.2       | 65.343668        | 0.967407      | 0.967552      |         |
| C=0.5, $\sigma=0.2$ , LR=0.005  | 0.005  | 0.50 | 0.2       | 65.343668        | 0.955556      | 0.954955      |         |
| C=1.0, $\sigma=0.2$ , LR=0.005  | 0.005  | 1.00 | 0.2       | 65.343668        | 0.944444      | 0.943139      |         |
| C=1.0, $\sigma=0.4$ , LR=0.005  | 0.005  | 1.00 | 0.4       | 13.803888        | 0.905926      | 0.904726      |         |
| C=3.31, $\sigma=0.4$ , LR=0.005 | 0.005  | 3.31 | 0.4       | 13.803888        | 0.901481      | 0.899471      |         |
| C=0.5, $\sigma=0.4$ , LR=0.005  | 0.005  | 0.50 | 0.4       | 13.803888        | 0.894815      | 0.899004      |         |
| C=0.5, $\sigma=0.8$ , LR=0.005  | 0.005  | 0.50 | 0.8       | 2.593935         | 0.777778      | 0.795082      |         |
| C=3.31, $\sigma=0.2$ , LR=0.001 | 0.001  | 3.31 | 0.2       | 65.343668        | 0.790370      | 0.766694      |         |
| C=1.0, $\sigma=0.8$ , LR=0.005  | 0.005  | 1.00 | 0.8       | 2.593935         | 0.761481      | 0.766667      |         |
| C=3.31, $\sigma=0.8$ , LR=0.005 | 0.005  | 3.31 | 0.8       | 2.593935         | 0.777778      | 0.760000      |         |
| C=0.5, $\sigma=0.2$ , LR=0.001  | 0.001  | 0.50 | 0.2       | 65.343668        | 0.756296      | 0.709107      |         |
| C=1.0, $\sigma=0.2$ , LR=0.001  | 0.001  | 1.00 | 0.2       | 65.343668        | 0.730370      | 0.654649      |         |
| C=3.31, $\sigma=0.4$ , LR=0.001 | 0.001  | 3.31 | 0.4       | 13.803888        | 0.673333      | 0.557673      |         |
| C=1.0, $\sigma=0.4$ , LR=0.001  | 0.001  | 1.00 | 0.4       | 13.803888        | 0.579259      | 0.445312      |         |
| C=3.31, $\sigma=0.8$ , LR=0.001 | 0.001  | 3.31 | 0.8       | 2.593935         | 0.508148      | 0.416520      |         |
| C=1.0, $\sigma=0.8$ , LR=0.001  | 0.001  | 1.00 | 0.8       | 2.593935         | 0.512593      | 0.412500      |         |
| C=0.5, $\sigma=0.8$ , LR=0.001  | 0.001  | 0.50 | 0.8       | 2.593935         | 0.512593      | 0.410394      |         |
| C=0.5, $\sigma=0.4$ , LR=0.001  | 0.001  | 0.50 | 0.4       | 13.803888        | 0.502963      | 0.407767      |         |

Plotting Top 5 DP-SGD Runs...

IndexError

Traceback (most recent call last)

/tmp/ipython-input-3258789571.py in <cell line: 0>()

294 run\_id = row['run\_id']

295 history\_df = all\_dp\_histories[run\_id]

296 ax1.plot(history\_df['epoch'], history\_df['accuracy'], label=f'{run\_id} (Accuracy)', color=colors[i])

297

298 ax1.set\_xlabel('Epoch')

IndexError: index 15 is out of bounds for axis 0 with size 5



Next steps: [Explain error](#)

```

if __name__ == '__main__':

    baseline_config = {'lr': 1e-3, 'batch_size': 128, 'epochs': 50, 'patience': 5}
    baseline_metrics, baseline_history = train_baseline_robust(baseline_config)

    DP_BATCH_SIZE = 256
    DP_EPOCHS = 15

    temp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
    temp_loader = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
    adaptive_c = find_adaptive_clip_norm(temp_model, temp_loader, cpu_device)

    CLIP_NORMS = sorted(list(set([0.5, 1.0, round(adaptive_c, 2)])))
    NOISE_MULTIPLIERS = [0.5, 1, 2]
    LEARNING_RATES = [5e-3]

    print(f"\nStarting grid sweep with C values: {CLIP_NORMS}")

    results = []
    all_dp_histories = {}

    for lr in LEARNING_RATES:
        for c in CLIP_NORMS:
            for sigma in NOISE_MULTIPLIERS:
                run_id = f"C={c}, σ={sigma}, LR={lr}"
                print("\n" + "-" * 50 + f"\nStarting DP-SGD Sweep Run: {run_id}")

                dp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
                dp_optimizer = AdamW(dp_model.parameters(), lr=lr)
                dp_criterion = nn.BCEWithLogitsLoss()

                train_loader_cpu = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
                test_loader_cpu = DataLoader(test_dataset, batch_size=DP_BATCH_SIZE, collate_fn=collate_bow)

                accountant = RDPAccountant()
                DELTA = 1 / len(train_dataset)
                sample_rate = DP_BATCH_SIZE / len(train_dataset)

                history = []
                for epoch in range(DP_EPOCHS):
                    train_manual_dp(dp_model, train_loader_cpu, dp_optimizer, dp_criterion, cpu_device, c, sigma)
                    accountant.step(noise_multiplier=sigma, sample_rate=sample_rate)
                    epsilon = accountant.get_epsilon(delta=DELTA)

                    epoch_metrics = eval_model(dp_model, test_loader_cpu, cpu_device)
                    history.append({
                        'epoch': epoch + 1,
                        'accuracy': epoch_metrics['accuracy'],
                        'epsilon': epsilon
                    })
                    print(f"Epoch {epoch+1}/{DP_EPOCHS} | Epsilon: {epsilon:.2f} | Test Accuracy: {epoch_metrics['accuracy']:.4f}")

                all_dp_histories[run_id] = pd.DataFrame(history)
                final_metrics = history[-1]

                results.append({
                    "run_id": run_id, "lr": lr, "clip_norm": c, "noise_multiplier": sigma,
                    "final_epsilon": final_metrics['epsilon'], "accuracy": final_metrics['accuracy'], "f1": eval_model(
                        dp_model, test_loader_cpu, cpu_device)
                })

    results_df = pd.DataFrame(results).rename(columns={"accuracy": "test_accuracy", "f1": "test_f1"})
    display_cols = ['run_id', 'lr', 'clip_norm', 'noise_multiplier', 'final_epsilon', 'test_accuracy', 'test_f1']
    results_df = results_df[display_cols].sort_values(by="test_f1", ascending=False)

    print("\n\n" + "=" * 80 + "\n\n          Final Tuned Sweep Results: Baseline vs. DP-SGD\n\n" + "=" * 80)
    print(f"BASELINE MODEL (Bag-of-Words, Non-Private):")
    print(f"Accuracy: {baseline_metrics['accuracy']:.4f}, F1 Score: {baseline_metrics['f1']:.4f}\n")
    print(f"DP-SGD MODELS (Bag-of-Words, Private, δ={DELTA:.2E}):")
    print(results_df.to_string(index=False))
    print("=" * 80)

    print("\n\n" + "=" * 80 + "\n\n          Plotting All DP-SGD Runs...\n\n" + "=" * 80)
    all_runs = results_df
    fig, ax1 = plt.subplots(figsize=(15, 9))

```

```

colors = plt.cm.viridis(np.linspace(0, 1, len(all_runs)))

ax1.plot(baseline_history['epoch'], baseline_history['val_accuracy'],
         label='Baseline (Val Accuracy)', color='black', linestyle='--', linewidth=2.5, zorder=100)

for i, (idx, row) in enumerate(all_runs.iterrows()):
    run_id = row['run_id']
    history_df = all_dp_histories[run_id]
    ax1.plot(history_df['epoch'], history_df['accuracy'], label=f'{run_id} (Accuracy)', color=colors[i])

ax1.set_xlabel('Epoch')
ax1.set_ylabel('Test Accuracy')
ax1.set_title('Model Accuracy and Privacy Loss ( $\epsilon$ ) vs. Epoch for All DP-SGD Runs')
ax1.grid(True, which='both', linestyle='--', linewidth=0.5)
ax1.set_ylim(bottom=0.45)

ax2 = ax1.twinx()
for i, (idx, row) in enumerate(all_runs.iterrows()):
    run_id = row['run_id']
    history_df = all_dp_histories[run_id]
    ax2.plot(history_df['epoch'], history_df['epsilon'], linestyle=':', marker='.', markersize=5, color=colors[i])

ax2.set_ylabel('Epsilon ( $\epsilon$ )')

lines, labels = ax1.get_legend_handles_labels()
ax1.legend(lines, labels, loc='center left', bbox_to_anchor=(1.1, 0.5), title="Runs")

fig.tight_layout()
plt.savefig("dp_sgd_all_learning_curves.png", bbox_inches='tight')
print("\nPlot saved to dp_sgd_all_learning_curves.png")
plt.show()

```



```
Using device for Baseline: cuda
Using device for DP-SGD: cpu
```

```
=====
Running Improved Non-Private Baseline Training...
=====
```

```
Epoch 1/50: 100%                               50/50 [00:00<00:00, 227.86it/s]
Epoch 1: Val F1: 0.8151, Current LR: 1.00E-03
Epoch 2/50: 100%                               50/50 [00:00<00:00, 258.28it/s]
Epoch 2: Val F1: 0.9419, Current LR: 1.00E-03
Epoch 3/50: 100%                               50/50 [00:00<00:00, 252.89it/s]
Epoch 3: Val F1: 0.9769, Current LR: 1.00E-03
Epoch 4/50: 100%                               50/50 [00:00<00:00, 262.61it/s]
Epoch 4: Val F1: 0.9971, Current LR: 1.00E-03
Epoch 5/50: 100%                               50/50 [00:00<00:00, 266.38it/s]
Epoch 5: Val F1: 0.9986, Current LR: 1.00E-03
Epoch 6/50: 100%                               50/50 [00:00<00:00, 208.72it/s]
Epoch 6: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 7/50: 100%                               50/50 [00:00<00:00, 256.35it/s]
Epoch 7: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 8/50: 100%                               50/50 [00:00<00:00, 243.97it/s]
Epoch 8: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 9/50: 100%                               50/50 [00:00<00:00, 245.55it/s]
Epoch 9: Val F1: 0.9986, Current LR: 1.00E-03
Epoch 10/50: 100%                              50/50 [00:00<00:00, 258.00it/s]
Epoch 10: Val F1: 0.9993, Current LR: 1.00E-03
Epoch 11/50: 100%                              50/50 [00:00<00:00, 235.53it/s]
Epoch 11: Val F1: 0.9993, Current LR: 1.00E-04
--- Early stopping triggered. ---
```

```
Loading best model and evaluating on the test set...
```

```
IMPROVED BASELINE RESULTS:
```

```
-> Final Test Accuracy: 1.0000
-> Final Test F1 Score: 1.0000
```

```
--- Running epoch to find adaptive clipping norm ---
```

```
Analyzing Gradients: 100%                        25/25 [00:07<00:00, 3.49it/s]
```

```
--- Found median gradient norm: 2.9134 ---
```

```
Starting grid sweep with C values: [0.5, 1.0, np.float64(2.91)]
```

```
-----
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =0.5, LR=0.005
```

```
Manual DP Batch: 100%                          25/25 [00:13<00:00, 1.98it/s]
Epoch 1/15 | Epsilon: 4.63 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.11it/s]
Epoch 2/15 | Epsilon: 5.29 | Test Accuracy: 0.5000
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.10it/s]
Epoch 3/15 | Epsilon: 5.71 | Test Accuracy: 0.5207
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.07it/s]
Epoch 4/15 | Epsilon: 6.04 | Test Accuracy: 0.5059
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.14it/s]
Epoch 5/15 | Epsilon: 6.32 | Test Accuracy: 0.6200
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.19it/s]
Epoch 6/15 | Epsilon: 6.56 | Test Accuracy: 0.7407
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.19it/s]
Epoch 7/15 | Epsilon: 6.81 | Test Accuracy: 0.5607
Manual DP Batch: 100%                          25/25 [00:13<00:00, 2.19it/s]
Epoch 8/15 | Epsilon: 6.99 | Test Accuracy: 0.6496
Manual DP Batch: 100%                          25/25 [00:14<00:00, 1.79it/s]
Epoch 9/15 | Epsilon: 7.17 | Test Accuracy: 0.7689
Manual DP Batch: 100%                          25/25 [00:13<00:00, 1.80it/s]
Epoch 10/15 | Epsilon: 7.36 | Test Accuracy: 0.7548
Manual DP Batch: 100%                          25/25 [00:13<00:00, 1.94it/s]
```

```

Epoch 11/15 | Epsilon: 7.54 | Test Accuracy: 0.7874
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.02it/s]
Epoch 12/15 | Epsilon: 7.70 | Test Accuracy: 0.8015
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.09it/s]
Epoch 13/15 | Epsilon: 7.84 | Test Accuracy: 0.8222
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.15it/s]
Epoch 14/15 | Epsilon: 7.98 | Test Accuracy: 0.8333
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.15it/s]
Epoch 15/15 | Epsilon: 8.12 | Test Accuracy: 0.8356

```

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =1, LR=0.005

```

Manual DP Batch: 100%                25/25 [00:13<00:00, 2.11it/s]
Epoch 1/15 | Epsilon: 1.04 | Test Accuracy: 0.5074
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.20it/s]
Epoch 2/15 | Epsilon: 1.11 | Test Accuracy: 0.5474
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.18it/s]
Epoch 3/15 | Epsilon: 1.17 | Test Accuracy: 0.5296
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.18it/s]
Epoch 4/15 | Epsilon: 1.21 | Test Accuracy: 0.5015
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.99it/s]
Epoch 5/15 | Epsilon: 1.24 | Test Accuracy: 0.5000
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.80it/s]
Epoch 6/15 | Epsilon: 1.28 | Test Accuracy: 0.5000
Manual DP Batch: 100%                25/25 [00:14<00:00, 1.99it/s]
Epoch 7/15 | Epsilon: 1.31 | Test Accuracy: 0.5548
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.07it/s]
Epoch 8/15 | Epsilon: 1.33 | Test Accuracy: 0.6044
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.13it/s]
Epoch 9/15 | Epsilon: 1.36 | Test Accuracy: 0.5815
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.13it/s]
Epoch 10/15 | Epsilon: 1.39 | Test Accuracy: 0.7030
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.02it/s]
Epoch 11/15 | Epsilon: 1.41 | Test Accuracy: 0.7607
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.16it/s]
Epoch 12/15 | Epsilon: 1.43 | Test Accuracy: 0.7274
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.18it/s]
Epoch 13/15 | Epsilon: 1.46 | Test Accuracy: 0.7644
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.18it/s]
Epoch 14/15 | Epsilon: 1.48 | Test Accuracy: 0.7519
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.94it/s]
Epoch 15/15 | Epsilon: 1.50 | Test Accuracy: 0.7659

```

-----  
Starting DP-SGD Sweep Run: C=0.5,  $\sigma$ =2, LR=0.005

```

Manual DP Batch: 100%                25/25 [00:13<00:00, 1.79it/s]
Epoch 1/15 | Epsilon: 0.20 | Test Accuracy: 0.4985
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.90it/s]
Epoch 2/15 | Epsilon: 0.21 | Test Accuracy: 0.5296
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.04it/s]
Epoch 3/15 | Epsilon: 0.22 | Test Accuracy: 0.4985
Manual DP Batch: 100%                25/25 [00:13<00:00, 1.86it/s]
Epoch 4/15 | Epsilon: 0.23 | Test Accuracy: 0.4985
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.14it/s]
Epoch 5/15 | Epsilon: 0.24 | Test Accuracy: 0.4985
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.18it/s]
Epoch 6/15 | Epsilon: 0.25 | Test Accuracy: 0.4985
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.14it/s]
Epoch 7/15 | Epsilon: 0.26 | Test Accuracy: 0.5778
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.16it/s]
Epoch 8/15 | Epsilon: 0.27 | Test Accuracy: 0.5252
Manual DP Batch: 100%                25/25 [00:13<00:00, 2.16it/s]

```

|                                                           |                               |
|-----------------------------------------------------------|-------------------------------|
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.19it/s] |
| Epoch 9/15   Epsilon: 0.28   Test Accuracy: 0.5637        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.99it/s] |
| Epoch 10/15   Epsilon: 0.28   Test Accuracy: 0.5674       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.81it/s] |
| Epoch 11/15   Epsilon: 0.29   Test Accuracy: 0.5237       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.87it/s] |
| Epoch 12/15   Epsilon: 0.30   Test Accuracy: 0.5422       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.00it/s] |
| Epoch 13/15   Epsilon: 0.31   Test Accuracy: 0.5222       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 14/15   Epsilon: 0.32   Test Accuracy: 0.5422       |                               |
| Manual DP Batch: 100%                                     |                               |
| Epoch 15/15   Epsilon: 0.33   Test Accuracy: 0.5674       |                               |
| -----                                                     |                               |
| Starting DP-SGD Sweep Run: C=1.0, $\sigma$ =0.5, LR=0.005 |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.93it/s] |
| Epoch 1/15   Epsilon: 4.63   Test Accuracy: 0.5059        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 2/15   Epsilon: 5.29   Test Accuracy: 0.5748        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.17it/s] |
| Epoch 3/15   Epsilon: 5.71   Test Accuracy: 0.5015        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.16it/s] |
| Epoch 4/15   Epsilon: 6.04   Test Accuracy: 0.5326        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.16it/s] |
| Epoch 5/15   Epsilon: 6.32   Test Accuracy: 0.5993        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.88it/s] |
| Epoch 6/15   Epsilon: 6.56   Test Accuracy: 0.6541        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.83it/s] |
| Epoch 7/15   Epsilon: 6.81   Test Accuracy: 0.8007        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.98it/s] |
| Epoch 8/15   Epsilon: 6.99   Test Accuracy: 0.8059        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.11it/s] |
| Epoch 9/15   Epsilon: 7.17   Test Accuracy: 0.7970        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 10/15   Epsilon: 7.36   Test Accuracy: 0.8096       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.15it/s] |
| Epoch 11/15   Epsilon: 7.54   Test Accuracy: 0.8148       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 12/15   Epsilon: 7.70   Test Accuracy: 0.8385       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.12it/s] |
| Epoch 13/15   Epsilon: 7.84   Test Accuracy: 0.8467       |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 2.14it/s] |
| Epoch 14/15   Epsilon: 7.98   Test Accuracy: 0.8519       |                               |
| Manual DP Batch: 100%                                     |                               |
| Epoch 15/15   Epsilon: 8.12   Test Accuracy: 0.8356       |                               |
| -----                                                     |                               |
| Starting DP-SGD Sweep Run: C=1.0, $\sigma$ =1, LR=0.005   |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:14<00:00, 1.79it/s] |
| Epoch 1/15   Epsilon: 1.04   Test Accuracy: 0.5607        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 1.95it/s] |
| Epoch 2/15   Epsilon: 1.11   Test Accuracy: 0.5081        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.05it/s] |
| Epoch 3/15   Epsilon: 1.17   Test Accuracy: 0.5281        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.09it/s] |
| Epoch 4/15   Epsilon: 1.21   Test Accuracy: 0.5200        |                               |
| Manual DP Batch: 100%                                     | 25/25 [00:13<00:00, 2.14it/s] |
| Epoch 5/15   Epsilon: 1.24   Test Accuracy: 0.5141        |                               |
| Manual DP Batch: 100%                                     |                               |
| Epoch 6/15   Epsilon: 1.28   Test Accuracy: 0.6059        |                               |



Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 7/15 | Epsilon: 1.31 | Test Accuracy: 0.5052

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 8/15 | Epsilon: 1.33 | Test Accuracy: 0.6030

Manual DP Batch: 100%25/25 [00:13<00:00, 2.16it/s]

Epoch 9/15 | Epsilon: 1.36 | Test Accuracy: 0.7044

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 10/15 | Epsilon: 1.39 | Test Accuracy: 0.7333

Manual DP Batch: 100%25/25 [00:13<00:00, 1.89it/s]

Epoch 11/15 | Epsilon: 1.41 | Test Accuracy: 0.7356

Manual DP Batch: 100%25/25 [00:13<00:00, 1.81it/s]

Epoch 12/15 | Epsilon: 1.43 | Test Accuracy: 0.7726

Manual DP Batch: 100%25/25 [00:14<00:00, 2.02it/s]

Epoch 13/15 | Epsilon: 1.46 | Test Accuracy: 0.7852

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 14/15 | Epsilon: 1.48 | Test Accuracy: 0.7941

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 15/15 | Epsilon: 1.50 | Test Accuracy: 0.7570

-----  
Starting DP-SGD Sweep Run: C=1.0,  $\sigma$ =2, LR=0.005

Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 1/15 | Epsilon: 0.20 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.13it/s]

Epoch 2/15 | Epsilon: 0.21 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.16it/s]

Epoch 3/15 | Epsilon: 0.22 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 4/15 | Epsilon: 0.23 | Test Accuracy: 0.4985

Manual DP Batch: 100%25/25 [00:13<00:00, 1.91it/s]

Epoch 5/15 | Epsilon: 0.24 | Test Accuracy: 0.5689

Manual DP Batch: 100%25/25 [00:14<00:00, 1.73it/s]

Epoch 6/15 | Epsilon: 0.25 | Test Accuracy: 0.5267

Manual DP Batch: 100%25/25 [00:13<00:00, 1.93it/s]

Epoch 7/15 | Epsilon: 0.26 | Test Accuracy: 0.5726

Manual DP Batch: 100%25/25 [00:13<00:00, 2.07it/s]

Epoch 8/15 | Epsilon: 0.27 | Test Accuracy: 0.5904

Manual DP Batch: 100%25/25 [00:13<00:00, 2.11it/s]

Epoch 9/15 | Epsilon: 0.28 | Test Accuracy: 0.5993

Manual DP Batch: 100%25/25 [00:13<00:00, 2.07it/s]

Epoch 10/15 | Epsilon: 0.28 | Test Accuracy: 0.5407

Manual DP Batch: 100%25/25 [00:14<00:00, 2.13it/s]

Epoch 11/15 | Epsilon: 0.29 | Test Accuracy: 0.5985

Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 12/15 | Epsilon: 0.30 | Test Accuracy: 0.5407

Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 13/15 | Epsilon: 0.31 | Test Accuracy: 0.5667

Manual DP Batch: 100%25/25 [00:13<00:00, 1.98it/s]

Epoch 14/15 | Epsilon: 0.32 | Test Accuracy: 0.6674

Manual DP Batch: 100%25/25 [00:14<00:00, 1.76it/s]

Epoch 15/15 | Epsilon: 0.33 | Test Accuracy: 0.7037

-----  
Starting DP-SGD Sweep Run: C=2.91,  $\sigma$ =0.5, LR=0.005

Manual DP Batch: 100%25/25 [00:13<00:00, 1.88it/s]

Epoch 1/15 | Epsilon: 4.63 | Test Accuracy: 0.6356

Manual DP Batch: 100%25/25 [00:13<00:00, 2.02it/s]

Epoch 2/15 | Epsilon: 5.29 | Test Accuracy: 0.5356

Manual DP Batch: 100%25/25 [00:13<00:00, 2.08it/s]

Epoch 3/15 | Epsilon: 5.71 | Test Accuracy: 0.7496

Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 4/15 | Epsilon: 6.04 | Test Accuracy: 0.6330

```
Epoch 4/15 | Epsilon: 0.04 | Test Accuracy: 0.0250
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.14it/s]

Epoch 5/15 | Epsilon: 6.32 | Test Accuracy: 0.6815
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.12it/s]

Epoch 6/15 | Epsilon: 6.56 | Test Accuracy: 0.7778
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.13it/s]

Epoch 7/15 | Epsilon: 6.81 | Test Accuracy: 0.7963
Manual DP Batch: 100%
25/25 [00:14<00:00, 1.58it/s]

Epoch 8/15 | Epsilon: 6.99 | Test Accuracy: 0.8326
Manual DP Batch: 100%
25/25 [00:14<00:00, 1.67it/s]

Epoch 9/15 | Epsilon: 7.17 | Test Accuracy: 0.8474
Manual DP Batch: 100%
25/25 [00:14<00:00, 1.92it/s]

Epoch 10/15 | Epsilon: 7.36 | Test Accuracy: 0.8667
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.07it/s]

Epoch 11/15 | Epsilon: 7.54 | Test Accuracy: 0.8570
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.08it/s]

Epoch 12/15 | Epsilon: 7.70 | Test Accuracy: 0.8815
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.10it/s]

Epoch 13/15 | Epsilon: 7.84 | Test Accuracy: 0.8726
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.11it/s]

Epoch 14/15 | Epsilon: 7.98 | Test Accuracy: 0.8696
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.10it/s]

Epoch 15/15 | Epsilon: 8.12 | Test Accuracy: 0.8956

-----
Starting DP-SGD Sweep Run: C=2.91, σ=1, LR=0.005
Manual DP Batch: 100%
25/25 [00:13<00:00, 1.99it/s]

Epoch 1/15 | Epsilon: 1.04 | Test Accuracy: 0.5000
Manual DP Batch: 100%
25/25 [00:14<00:00, 1.76it/s]

Epoch 2/15 | Epsilon: 1.11 | Test Accuracy: 0.5296
Manual DP Batch: 100%
25/25 [00:13<00:00, 1.89it/s]

Epoch 3/15 | Epsilon: 1.17 | Test Accuracy: 0.5059
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.00it/s]

Epoch 4/15 | Epsilon: 1.21 | Test Accuracy: 0.6274
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.11it/s]

Epoch 5/15 | Epsilon: 1.24 | Test Accuracy: 0.6111
Manual DP Batch: 100%
25/25 [00:14<00:00, 2.04it/s]

Epoch 6/15 | Epsilon: 1.28 | Test Accuracy: 0.6852
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.10it/s]

Epoch 7/15 | Epsilon: 1.31 | Test Accuracy: 0.7185
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.12it/s]

Epoch 8/15 | Epsilon: 1.33 | Test Accuracy: 0.7526
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.12it/s]

Epoch 9/15 | Epsilon: 1.36 | Test Accuracy: 0.7274
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.10it/s]

Epoch 10/15 | Epsilon: 1.39 | Test Accuracy: 0.7437
Manual DP Batch: 100%
25/25 [00:14<00:00, 1.76it/s]

Epoch 11/15 | Epsilon: 1.41 | Test Accuracy: 0.7652
Manual DP Batch: 100%
25/25 [00:13<00:00, 1.88it/s]

Epoch 12/15 | Epsilon: 1.43 | Test Accuracy: 0.7674
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.01it/s]

Epoch 13/15 | Epsilon: 1.46 | Test Accuracy: 0.8044
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.07it/s]

Epoch 14/15 | Epsilon: 1.48 | Test Accuracy: 0.8400
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.10it/s]

Epoch 15/15 | Epsilon: 1.50 | Test Accuracy: 0.8044

-----
Starting DP-SGD Sweep Run: C=2.91, σ=2, LR=0.005
Manual DP Batch: 100%
25/25 [00:13<00:00, 2.11it/s]

Epoch 1/15 | Epsilon: 0.20 | Test Accuracy: 0.5148
Manual DP Batch: 100%
25/25 [00:14<00:00, 2.10it/s]
```

Epoch 2/15 | Epsilon: 0.21 | Test Accuracy: 0.5000  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.10it/s]

Epoch 3/15 | Epsilon: 0.22 | Test Accuracy: 0.5052  
Manual DP Batch: 100%25/25 [00:14<00:00, 1.81it/s]

Epoch 4/15 | Epsilon: 0.23 | Test Accuracy: 0.5022  
Manual DP Batch: 100%25/25 [00:14<00:00, 1.79it/s]

Epoch 5/15 | Epsilon: 0.24 | Test Accuracy: 0.5000  
Manual DP Batch: 100%25/25 [00:13<00:00, 1.96it/s]

Epoch 6/15 | Epsilon: 0.25 | Test Accuracy: 0.5030  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.01it/s]

Epoch 7/15 | Epsilon: 0.26 | Test Accuracy: 0.5037  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.09it/s]

Epoch 8/15 | Epsilon: 0.27 | Test Accuracy: 0.5067  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 9/15 | Epsilon: 0.28 | Test Accuracy: 0.5978  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.12it/s]

Epoch 10/15 | Epsilon: 0.28 | Test Accuracy: 0.6526  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.14it/s]

Epoch 11/15 | Epsilon: 0.29 | Test Accuracy: 0.6815  
Manual DP Batch: 100%25/25 [00:13<00:00, 2.15it/s]

Epoch 12/15 | Epsilon: 0.30 | Test Accuracy: 0.6763  
Manual DP Batch: 100%25/25 [00:14<00:00, 1.79it/s]

Epoch 13/15 | Epsilon: 0.31 | Test Accuracy: 0.7037  
Manual DP Batch: 100%25/25 [00:14<00:00, 1.70it/s]

Epoch 14/15 | Epsilon: 0.32 | Test Accuracy: 0.6926  
Manual DP Batch: 100%25/25 [00:13<00:00, 1.89it/s]

Epoch 15/15 | Epsilon: 0.33 | Test Accuracy: 0.7259

=====

Final Tuned Sweep Results: Baseline vs. DP-SGD

=====

BASELINE MODEL (Bag-of-Words, Non-Private):

Accuracy: 1.0000, F1 Score: 1.0000

DP-SGD MODELS (Bag-of-Words, Private,  $\delta \approx 1.59E-04$ ):

|                                 | run_id | lr   | clip_norm | noise_multiplier | final_epsilon | test_accuracy | test_f1 |
|---------------------------------|--------|------|-----------|------------------|---------------|---------------|---------|
| C=2.91, $\sigma=0.5$ , LR=0.005 | 0.005  | 2.91 | 0.5       | 8.119425         | 0.895556      | 0.894855      |         |
| C=0.5, $\sigma=0.5$ , LR=0.005  | 0.005  | 0.50 | 0.5       | 8.119425         | 0.835556      | 0.830275      |         |
| C=1.0, $\sigma=0.5$ , LR=0.005  | 0.005  | 1.00 | 0.5       | 8.119425         | 0.835556      | 0.823248      |         |
| C=0.5, $\sigma=1$ , LR=0.005    | 0.005  | 0.50 | 1.0       | 1.499881         | 0.765926      | 0.777465      |         |
| C=2.91, $\sigma=1$ , LR=0.005   | 0.005  | 2.91 | 1.0       | 1.499881         | 0.804444      | 0.773973      |         |
| C=2.91, $\sigma=2$ , LR=0.005   | 0.005  | 2.91 | 2.0       | 0.326025         | 0.725926      | 0.719272      |         |
| C=1.0, $\sigma=1$ , LR=0.005    | 0.005  | 1.00 | 1.0       | 1.499881         | 0.757037      | 0.710247      |         |
| C=1.0, $\sigma=2$ , LR=0.005    | 0.005  | 1.00 | 2.0       | 0.326025         | 0.703704      | 0.689441      |         |
| C=0.5, $\sigma=2$ , LR=0.005    | 0.005  | 0.50 | 2.0       | 0.326025         | 0.567407      | 0.470054      |         |

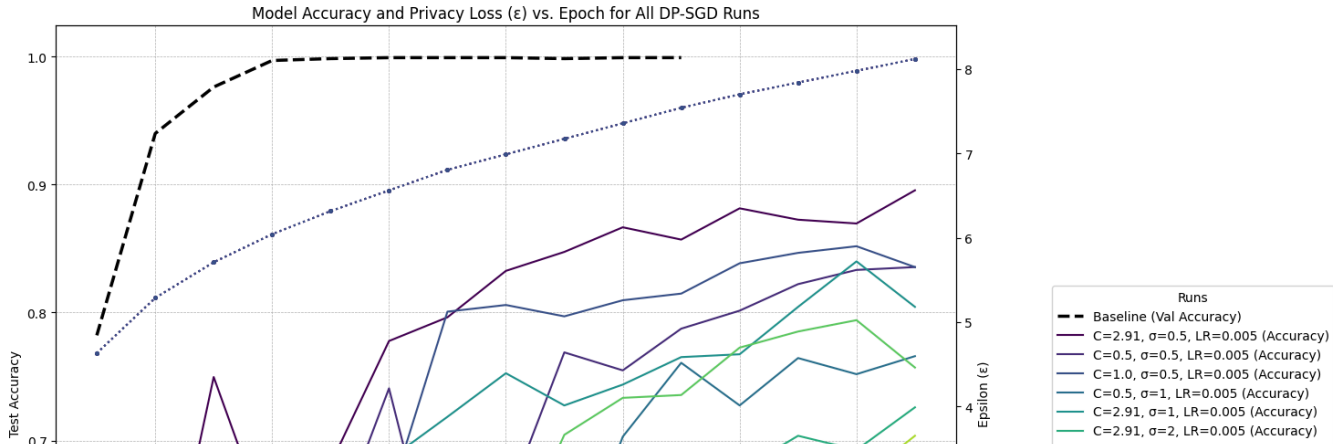
=====

=====

Plotting All DP-SGD Runs...

=====

Plot saved to dp\_sgd\_all\_learning\_curves.png



```

from sklearn.linear_model import LogisticRegression

# MIA with Logistic Regression Model
def compute_losses(model, loader, device):
    model.eval()
    criterion = nn.BCEWithLogitsLoss(reduction='none')
    losses = []
    with torch.no_grad():
        for texts, offsets, cats, labels in loader:
            texts, offsets, cats, labels = texts.to(device), offsets.to(device), cats.to(device), labels.to(device)
            logits = model(texts, offsets, cats)
            loss = criterion(logits, labels)
            losses.append(loss.cpu().numpy())
    return np.concatenate(losses)

def run_mia(target_model, train_loader, test_loader, device):
    member_losses = compute_losses(target_model, train_loader, device)
    non_member_losses = compute_losses(target_model, test_loader, device)

    X_attack = np.concatenate([member_losses, non_member_losses]).reshape(-1, 1)
    y_attack = np.concatenate([np.ones_like(member_losses), np.zeros_like(non_member_losses)])

    attack_model = LogisticRegression(class_weight='balanced')
    attack_model.fit(X_attack, y_attack)

    attack_accuracy = attack_model.score(X_attack, y_attack)
    return attack_accuracy

if __name__ == '__main__':

    # Baseline Model
    baseline_config = {'lr': 1e-3, 'batch_size': 128, 'epochs': 20, 'patience': 5}
    baseline_model, baseline_metrics = train_baseline_robust(baseline_config)

    # Using the best hyperparameters from previous sweep
    DP_BATCH_SIZE = 256
    DP_EPOCHS = 15
    DP_LR = 5e-3
    DP_NOISE_MULTIPLIER = 0.5
    DP_CLIP_NORM = 2.91

    print("\n" + "=" * 50 + f"\nTraining Best DP-SGD Model for MIA Comparison...\n" + "=" * 50)

    dp_model = BagOfWordsClassifier(VOCAB_SIZE, len(job_categories)).to(cpu_device)
    dp_optimizer = AdamW(dp_model.parameters(), lr=DP_LR)
    dp_criterion = nn.BCEWithLogitsLoss()

    train_loader_cpu = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=True, collate_fn=collate_bow)
    test_loader_cpu = DataLoader(test_dataset, batch_size=DP_BATCH_SIZE, collate_fn=collate_bow)

    for epoch in range(DP_EPOCHS):
        train_manual_dp(dp_model, train_loader_cpu, dp_optimizer, dp_criterion, cpu_device, DP_CLIP_NORM, DP_NOISE_MUI)
        print(f"DP Training Epoch {epoch+1}/{DP_EPOCHS} complete.")

    print("\nEvaluating final DP model...")
    dp_metrics = eval_model(dp_model, test_loader_cpu, cpu_device)
    print(f" -> Final DP Test Accuracy: {dp_metrics['accuracy']:.4f}\n -> Final DP Test F1 Score: {dp_metrics['f1']}")

    # Run and Compare MIAs
    print("\n" + "\nRunning Membership Inference Attacks...\n")

    # For equal weighting, we don't shuffle
    mia_train_loader_gpu = DataLoader(train_dataset, batch_size=baseline_config['batch_size'], shuffle=False, collate_fn=collate_bow)
    mia_test_loader_gpu = DataLoader(test_dataset, batch_size=baseline_config['batch_size'], shuffle=False, collate_fn=collate_bow)

    mia_train_loader_cpu = DataLoader(train_dataset, batch_size=DP_BATCH_SIZE, shuffle=False, collate_fn=collate_bow)
    mia_test_loader_cpu = DataLoader(test_dataset, batch_size=DP_BATCH_SIZE, shuffle=False, collate_fn=collate_bow)

    # Attack the baseline model
    baseline_mia_accuracy = run_mia(baseline_model, mia_train_loader_gpu, mia_test_loader_gpu, gpu_device)

    # Attack the DP model
    dp_mia_accuracy = run_mia(dp_model, mia_train_loader_cpu, mia_test_loader_cpu, cpu_device)

    # --- Final Report ---
    print("\n\n" + "=" * 70 + "\n          Final Report: Model Performance and MIA Vulnerability\n" + "=" * 70)

```

```
print("--- BASELINE MODEL (Non-Private) ---")
print(f"  Utility:      Accuracy={baseline_metrics['accuracy']:.4f}, F1 Score={baseline_metrics['f1']:.4f}")
print(f"  Vulnerability: MIA Accuracy={baseline_mia_accuracy:.4f}\n")

print("--- DP-SGD MODEL (Private) ---")
print(f"  Utility:      Accuracy={dp_metrics['accuracy']:.4f}, F1 Score={dp_metrics['f1']:.4f}")
print(f"  Vulnerability: MIA Accuracy={dp_mia_accuracy:.4f}\n")

print("--- Takeaway ---")
print("The non-private baseline model is highly vulnerable to Membership Inference Attacks, as shown by the high i")
print("The DP-SGD model is slightly against this attack,")
print("which shows the effectiveness of DP-SGD. To get lower attack accuracy, the noise multiplier has to be ramped")

Using device for Baseline: cuda
Using device for DP-SGD: cpu

=====
Running Non-Private Baseline Training...
```